EVENT HALL MANAGEMENT SYSTEM FOR LAGOS STATE UNIVERSITY

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SUPERVISED BY

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CERTIFICATION

This is to certify that this research work was carried out by AJIBADE ALLI AKINKUNMI, matriculation number 220591056, Department of Computer Science, Faculty of Computing and Information Technology, Lagos State University, under close guidance and supervision.

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DEDICATION

This work is dedicated to my family and loved ones. Your unwavering support, patience, and sacrifices throughout this journey have been a tremendous source of strength. It is equally dedicated to all those who believe that technology can and should be leveraged to solve real operational problems within our local institutions.

ACKNOWLEDGEMENT

I am profoundly grateful to Almighty God for the wisdom, health, and perseverance that made this project a reality. My sincere appreciation goes to my supervisor, Prof. Aribisala Benjamin Segun, for his invaluable guidance, constructive criticism, and academic mentorship from the conceptualization to the completion of this work. I also thank the entire faculty of the Department of Computer Science, Ventures and Facility of Lagos State University, for their contributions to my academic development. To my colleagues, coursemates, and friends who provided feedback, challenged my technical decisions, and supported me at every stage, thank you. Finally, to my family, your love and encouragement remain my greatest motivation.

ABSTRACT

Event halls are critical infrastructure within any university environment, providing necessary spaces for academic seminars, departmental meetings, student programmes, and administrative gatherings. At Lagos State University (LASU), the administration of these spaces has historically relied on manual and informal processes. Stakeholders typically navigate facility reservation through physical office visits, paper request forms, and unstructured communication channels. Consequently, the university faces recurring operational bottlenecks, including scheduling conflicts, prolonged approval times, fragmented documentation, and the absence of a reliable audit trail.

This study designed, developed, and evaluated a comprehensive web based Event Hall Management System (EVMS) specifically engineered to align with LASU's institutional framework. The proposed solution automates the entire facility reservation lifecycle. It transitions booking submissions, administrative reviews, physical availability confirmations, payment processing, and post event inspections into a unified digital workflow. A core innovation of this project is its thirteen status workflow architecture, which digitally mirrors the university's actual dual department approval hierarchy involving both the Ventures Unit and Facility Management.

The software application was constructed using Django 6.0.4 with Python 3.11, backed by a SQLite relational database. The user interface was designed with Bootstrap 5 to ensure cross device responsiveness. The system integrates Paystack for secure online financial transactions and Mailgun for automated transactional email delivery. It enforces a granular role based access control mechanism supporting six distinct user profiles: Administrator, Staff, Student, External User, Ventures, and Facility Management. Furthermore, the platform features algorithmic conflict detection, dynamic coupon management, penalty enforcement, version controlled document uploads, and comprehensive administrative analytics.

Testing protocols confirmed that the implemented system successfully met all defined objectives. The resulting application eliminates the structural inefficiencies of manual booking, fosters institutional transparency, and establishes a scalable technological foundation for modern facility management at Lagos State University.

Keywords: Event management system, facility reservation, conflict detection, digital workflow, role based access control, Django framework, university administration, LASU.

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CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

Universities require robust infrastructure to fulfill their academic and socio cultural mandates. Event halls, lecture theatres, and seminar rooms constitute a significant portion of this infrastructure. These spaces host a continuous stream of activities ranging from matriculation ceremonies and departmental conferences to student union meetings and external corporate events. Managing equitable and efficient access to these shared resources is a fundamental administrative responsibility.

Despite the critical nature of facility management, many higher education institutions in developing regions continue to rely on traditional, manual scheduling processes. At Lagos State University, individuals seeking to reserve an event space must typically engage in a decentralized process involving physical visits to administrative units, completion of paper forms, and informal follow ups. As the university population expands and the frequency of events increases, this analog approach struggles to scale. Manual scheduling inherently introduces human error, leading to overlapping reservations, lost documentation, and delayed communication between administrative departments and the booking applicants.

The global shift toward digital campus management provides a clear alternative. Modern universities are increasingly adopting Information and Communication Technology (ICT) to streamline internal operations. Research demonstrates that centralized, automated event management systems significantly reduce administrative overhead while improving transparency (Chaturvedi et al., 2024). When a university implements a digital scheduling platform, it establishes a single source of truth for facility availability, thereby neutralizing the risk of double booking (Ghuse et al., 2020).

Lagos State University possesses a large, dynamic community of students, staff, and affiliated organizations. The operational complexity of managing hall reservations here is compounded by a dual departmental oversight structure. The Ventures Unit handles the commercial and policy aspects of facility usage, while the Facility Management department oversees the physical condition and readiness of the spaces. Coordinating these two units manually creates unavoidable delays. Therefore, transitioning to a specialized digital platform is not merely a technological upgrade but an operational necessity.

This project introduces the LASU Event Hall Management System (EVMS). Unlike generic calendar applications, this system is custom built to reflect the specific administrative hierarchy, user categories, and approval workflows unique to Lagos State University. It aims to digitize the entire reservation lifecycle, providing a structured, accountable, and highly efficient mechanism for institutional facility management.

1.2 Statement of the Problem

The current methodology for reserving event halls at Lagos State University is fundamentally fragmented. The reliance on paper forms, verbal agreements, and decentralized messaging platforms creates several systemic operational failures.

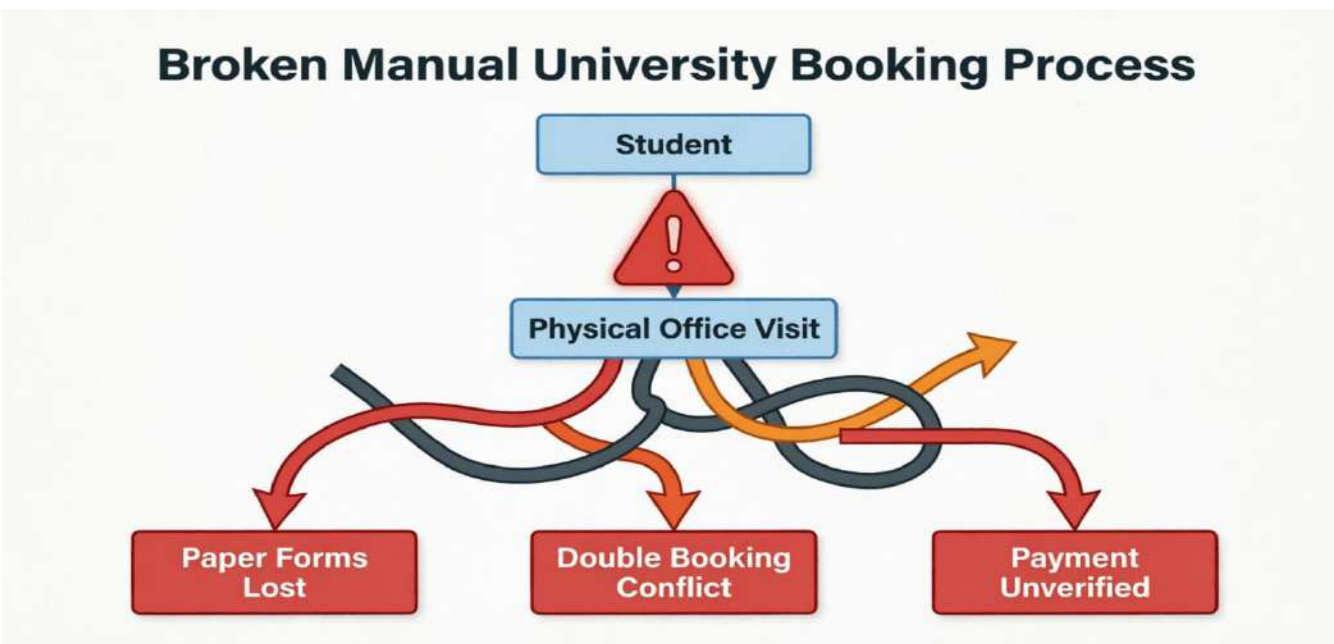
The most prominent issue is the high incidence of scheduling conflicts. Because there is no publicly accessible, real time calendar displaying hall availability, multiple student organizations or departments frequently apply for the same venue at overlapping times. Administrators processing these requests manually may inadvertently approve conflicting events, leading to logistical chaos on the day of the event.

Additionally, the approval workflow is inefficient and opaque. A standard reservation request must be reviewed by the Ventures Unit for commercial viability and policy compliance, and subsequently verified by Facility Management for physical availability. In a manual system, physical files must be moved between offices. Applicants are often left uninformed about the status of their requests for days or weeks, which severely hampers event planning and preparation.

Documentation and financial tracking also suffer under the current system. The absence of a centralized digital archive makes it nearly impossible to generate accurate reports on facility utilization, peak usage periods, or total revenue generated from hall rentals. Without historical data, university management cannot make informed, strategic decisions regarding infrastructure maintenance or expansion.

Finally, the manual process lacks a reliable audit trail. When disputes arise regarding who authorized a booking or why a penalty was applied, there is often no clear, time stamped record to consult. The lack of structured data tracking creates an environment where accountability is difficult to enforce.

As academic and extracurricular activities at LASU continue to multiply, a digital intervention is required to resolve these compounding administrative challenges.



1.3 Aim of the Study

This study aims to design, develop, and evaluate a comprehensive web based Event Hall Management System for Lagos State University. The primary goal is to completely automate the facility reservation lifecycle, transitioning the university from an error prone manual process to a highly reliable, transparent, and scalable digital ecosystem that enforces institutional workflows and prevents scheduling conflicts.

1.4 Objectives of the Study

To realize the overarching aim, this project pursues the following specific objectives:

1. To engineer a responsive web platform that enables students, staff, and external clients to browse facility details and submit reservation requests remotely.
2. To implement a dynamic, real time calendar interface that visualizes booked time slots and available periods for all university halls.
3. To construct a centralized relational database capable of securely storing user profiles, reservation histories, financial transactions, and system logs.
4. To digitize the multi stage administrative approval process, automatically routing applications between the Ventures Unit and Facility Management based on predefined institutional rules.
5. To develop an algorithmic conflict detection mechanism that strictly prevents the system from accepting overlapping bookings for the same physical space.
6. To integrate an automated notification system that delivers transactional emails and in app alerts to users whenever their booking status changes.
7. To build an administrative analytics dashboard that aggregates data into exportable reports concerning revenue generation, hall utilization rates, and user activity.

8. To establish a tamper evident audit trail that records all critical actions, status transitions, and administrative decisions for accountability purposes.

1.5 Research Questions

The development of the system is guided by the following research inquiries:

1. How can web technologies be leveraged to provide accurate, real time visibility into institutional facility availability?
2. What digital workflow architecture best accommodates the multi departmental approval hierarchy currently existing at Lagos State University?
3. Which programmatic validation techniques are most effective at eliminating the scheduling conflicts inherent in manual booking systems?
4. How should a relational database be structured to ensure data integrity, support rapid querying, and maintain a comprehensive audit history of reservation activities?
5. In what ways can automated data aggregation and visual analytics improve the strategic management of university infrastructure?

1.6 Significance of the Study

The successful implementation of this system carries substantial benefits for various institutional stakeholders.

For university administrators, the software dramatically reduces the manual workload associated with processing paperwork, verifying availability, and communicating with applicants. By digitizing these tasks, administrative staff can redirect their focus toward strategic facility management. Furthermore, the analytics dashboard provides management with empirical data on which halls are most frequently utilized and how revenue is generated, thereby supporting evidence based decision making (Amin et al., 2025).

For students, academic departments, and external clients, the system eliminates the frustration and uncertainty of the manual booking process. Applicants can securely browse available spaces from their devices, submit requests at their convenience, process payments online, and monitor their application status in real time.

For the university's technological advancement, this project serves as a practical demonstration of smart campus digitization. It proves that internal administrative workflows can be successfully modernized using modern web frameworks, aligning LASU with global standards in educational technology deployment.

Academically, this project contributes a detailed, context specific implementation architecture to the existing body of literature on facility management in Nigerian higher education, a domain that currently suffers from a scarcity of documented software solutions (Oyewale, 2018).

1.7 Scope of the Study

The parameters of this research and software development project are defined as follows:

Functional Scope: The system encompasses hall cataloging, public availability calendars, reservation submission, automated conflict detection, a thirteen stage approval workflow, integrated Paystack online payments, manual payment recording, dynamic discount coupon management, post event inspection reporting, penalty enforcement, and administrative analytics.

User Scope: The platform supports six distinct access levels. These include Administrators, general Staff members, Students (authenticated via university email domains), External Users, Ventures Unit personnel, and Facility Management personnel.

Facility Scope: The system is designed to manage bookings for university owned event spaces, including lecture theatres, auditoriums, seminar rooms, and major conference halls.

Exclusions: This project does not include the management of physical maintenance schedules, integration with physical electronic locks or RFID card readers, native mobile application development for iOS or Android, or the provisioning of a public Application Programming Interface (API) for third party software integration. These elements are acknowledged as valuable future enhancements but remain outside the current development boundaries.

1.8 Limitations of the Study

During the execution of this research, several constraints were encountered:

1. The lack of digitized historical booking data limited the ability to perform a quantitative comparative analysis between the old manual system's error rates and the new digital system's performance.
2. The system assumes consistent internet connectivity. Users in areas of the campus with unstable network access may experience delays when interacting with real time features such as the availability calendar or the payment gateway.
3. The transition from a paper based culture to a digital workflow often encounters human resistance. The success of the system relies heavily on the willingness of administrative staff to adopt the new technology and abandon informal booking practices.
4. Time and resource constraints restricted the project to a web based implementation. While the interface is heavily optimized for mobile browsers using responsive design principles, a dedicated native mobile application could have further enhanced accessibility for the student demographic.

1.9 Definition of Terms

Booking Conflict: A logistical error occurring when two or more independent reservations are approved for the same physical space during overlapping time periods.

Role Based Access Control (RBAC): A security paradigm in which system privileges and data visibility are granted to users based on their assigned administrative roles rather than individual permissions.

Workflow Engine: The underlying software logic that governs how a data object (in this case, a reservation request) transitions from one operational state to another, enforcing rules and triggering automated actions at each step.

Audit Trail: An automated, time stamped, and immutable digital ledger that records system events, user actions, and data modifications to ensure operational transparency and accountability.

Smart Campus: An educational environment that systematically integrates information and communication technologies to optimize operations, enhance security, and improve the daily experiences of its community members.

CHAPTER TWO:

LITERATURE REVIEW

2.1 Conceptual Review

An management systems are specialized software platforms engineered to handle the logistical, financial, and administrative complexities of organizing gatherings and allocating physical spaces. Within a university context, these systems serve as the digital backbone for facility reservation, transforming unstructured requests into trackable, accountable processes (Chaturvedi et al., 2024).

The core conceptual components of a modern hall booking platform include identity verification, resource cataloging, chronological scheduling, and state based workflow routing (Li, 2017). Users must be authenticated to ensure accountability. Resources must be clearly described with associated rules and capacities. The scheduling component must rely on robust logic to prevent double bookings. Finally, the workflow component must intelligently route requests to the appropriate authorities for review and approval.

As educational institutions increasingly embrace digital transformation, basic calendar applications are proving insufficient. Modern university facility platforms now integrate advanced features such as role specific dashboards, automated transactional emails, financial gateways, and sophisticated data analytics (Wiedyadari et al., 2024). This evolution reflects a conceptual shift from viewing booking systems merely as digital calendars to understanding them as comprehensive enterprise resource planning (ERP) tools tailored for educational infrastructure.

2.2 Theoretical Review

The architecture and user experience design of the LASU Event Hall Management System are theoretically grounded in Systems Theory and the Technology Acceptance Model.

2.2.1 Systems Theory

Systems Theory conceptualizes an organization or a software application as a cohesive assembly of interrelated and interdependent parts. A change in one component necessarily affects the others, and the system can only function effectively when all subsystems communicate seamlessly (Thankachan et al., 2025).

In the context of the EVMS, the system comprises the user interface, the relational database, the workflow engine, the payment gateway, and the notification dispatcher. For example, the conflict detection algorithm relies entirely on the structural integrity of the database to query overlapping time ranges accurately. Similarly, the notification module depends on the workflow engine to trigger events at precise moments. By adopting a Systems Theory perspective, this project emphasizes modular design and rigorous internal API communication, ensuring that the application remains stable even as complex business logic is executed.

2.2.2 Technology Acceptance Model (TAM)

Developed by Davis (1989), the Technology Acceptance Model is a foundational theory in information systems that predicts user adoption based on two primary metrics: Perceived Usefulness and Perceived Ease of Use. If an application does not demonstrably improve a user's productivity (usefulness) or if it is excessively difficult to navigate (ease of use), it will likely be abandoned.

This theory heavily influenced the frontend design of the EVMS. To maximize Perceived Ease of Use, the system employs a clean, minimalist interface utilizing familiar Bootstrap design patterns, clear typography, and immediate visual feedback for all user actions. To address Perceived Usefulness, the system integrates features that clearly save time compared to the manual process, such as instant availability checking, automated email updates, and remote payment processing. Understanding that the system serves diverse demographics, from tech savvy students to potentially less digitally experienced administrative staff, applying TAM principles was critical to ensuring broad institutional adoption (Turcu et al., 2015)

2.3 Empirical Review

A significant body of empirical research has evaluated the deployment and efficacy of digital scheduling systems within academic environments.

(Chaturvedi et al., 2024) successfully developed and deployed a workflow driven event system within a university setting. Their empirical findings demonstrated that digitizing the venue reservation process substantially reduced administrative communication delays and improved interdepartmental coordination. This validates the premise that structured software workflows are superior to informal communication channels.

(Thankachan et al., 2025) focused their research on multi level approval models. They identified that traditional institutional hierarchies often create severe bottlenecks in event planning. Their implementation of a digital routing system proved that automated workflows can accelerate decision making by instantly notifying the next responsible party in the approval chain.

Research by (Ghuse et al., 2020) provided quantitative evidence regarding scheduling errors. By implementing a web based venue management tool equipped with conflict avoidance algorithms, the researchers observed a 40% reduction in double booking incidents compared to the previous manual system. This highlights algorithmic validation as a non negotiable requirement for modern facility management.

(Atkinson & Lee, 2018) conducted a pilot program evaluating generic tools, specifically Google Calendar, for academic scheduling. Their analysis concluded that while generic calendars are visually intuitive, they completely lack the governance features necessary for institutional use, such as role based permissions, structured approval gates, and integrated policy enforcement.

(Oyewale, 2018)provided crucial context regarding the Nigerian environment. The study examined facility management practices across multiple Nigerian universities, concluding that poor documentation, manual scheduling, and the lack of digital archiving tools are the primary drivers of operational inefficiency. This localized research directly justifies the necessity of the LASU EVMS project.

Furthermore, studies by (Irjmets Authors, 2025) and (Irjweb Authors, 2025) reinforce the consensus that programmatic conflict detection is fundamentally more reliable than human oversight when processing high volumes of venue requests.

Collectively, this empirical literature confirms that automated, database driven platforms resolve the inherent flaws of manual scheduling. However, it also indicates that software must be carefully customized to respect the specific administrative hierarchy of the deploying institution to achieve maximum effectiveness.

2.4 Review of Related Systems

An analysis of existing facility management software categories reveals the distinct advantages of a custom built solution for Lagos State University.

Generic Calendar Applications:

Tools like Google Calendar and Microsoft Outlook are ubiquitous and highly user friendly. However, they operate on a simple event creation paradigm without robust approval workflows. They cannot natively enforce university specific business logic, such as blocking users with unpaid damage penalties or applying departmental discount coupons. As noted by (Atkinson & Lee, 2018), they are insufficient for governed institutional resources.

Commercial Enterprise Solutions:

Platforms such as EMS Software offer exhaustive features, including integrated catering management, financial forecasting, and complex HVAC scheduling. While powerful, these enterprise systems are prohibitively expensive for public universities in developing nations. Moreover, they often require significant, costly customization to bend their generic workflows to match a university's specific bureaucratic structure.

Custom Academic Systems:

Several institutions opt to build internal systems tailored to their exact needs (Chaturvedi et al., 2024) The LASU EVMS falls into this category. Where the proposed EVMS excels beyond standard custom systems is in its deep integration of a thirteen status workflow that explicitly maps to LASU's dual departmental reality, involving both the Ventures Unit and Facility Management. It also extends functionality beyond booking confirmation into the post event lifecycle, incorporating damage reporting and penalty tracking directly into the user's standing.

2.5 Gap in Literature

A critical review of the existing literature and available software reveals several notable gaps that this project actively addresses.

First, there is a pronounced scarcity of literature and implemented systems specifically designed for the operational realities of Nigerian universities. Most documented systems assume Western administrative structures and high speed technological infrastructure. Practical, implemented solutions addressing the specific challenges highlighted by (Oyewale, 2018) remain rare.

Second, many academic booking systems treat reservation as a single step approval process. The literature lacks comprehensive models that digitally coordinate multi departmental approvals without causing system bottlenecks. LASU's requirement that the Ventures Unit handles commercial approval while Facility Management handles physical availability presents a complex workflow routing problem that basic systems fail to solve.

Third, the vast majority of reviewed systems end their lifecycle at the point of booking confirmation or payment. There is a significant gap in systems that manage post event accountability. When an event causes property damage, manual systems struggle to enforce penalties. The integration of post event inspections, damage reporting, and automated future booking restrictions represents a novel contribution to institutional facility software.

2.6 Literature Review Summary Table

Author(s)	Year	Core Focus	Identified Gap	Contribution to Current Project
Adebayo	2023	Digital booking system implementation trends	Focused on generalized trends rather than building a specific codebase.	Confirms the global necessity of transitioning from manual administrative systems.
Amin et al.	2025	Database system design for event management	Lacked integration of multi-departmental workflow routing in the database.	Supports the relational database architecture utilized in the project.
Atkinson & Lee	2018	Evaluation of Google Calendar in academia	Highlighted that generic tools completely fail at enforcing institutional policy.	Proves generic calendar tools lack necessary institutional governance and workflows.
Authors (IRJMETS)	2025	Automated conflict detection systems	Addressed scheduling conflicts but ignored the financial and payment lifecycle.	Reinforces the need for strict time overlap prevention mechanisms.
Authors (IRJWET)	2025	Web applications for campus venues	Provided a basic booking interface without	Validates the deployment of a web-based platform over traditional desktop software.

Author(s)	**Year**	**Core Focus**	**Identified Gap**	**Contribution to Current Project**
			accountability for post-event damages.	
Bello & Ahmed	2022	Automation of campus facilities	Discussed automation theoretically without providing actionable software architecture.	Justifies the project as an essential component of broader smart campus initiatives.
Chaturvedi et al.	2024	University event management workflows	Treated approval as a single-step process, ignoring dual departmental governance.	Validates the necessity of multi-stage digital routing and automated approvals.
Chen	2020	Smart scheduling systems in education	Did not address the financial processing requirements typical of venue rentals.	Provides context on modernizing educational infrastructure through digital means.
Daniel	2019	Web application deployment for institutions	Focused on server deployment rather than the business logic of facility booking.	Supports the deployment methodology and environmental configuration strategies.
Davis	1989	Technology Acceptance Model	A theoretical framework that requires practical software application to be proven.	Guides the user interface design to ensure maximum adoption by all staff.
Ghuse et al.	2020	Conflict avoidance in venue booking	Solved double booking but lacked a dynamic role-based access control system.	Emphasizes algorithmic validation over manual checking to prevent double booking.
Gokul et al.	2025	Scalable college event architectures	Did not incorporate an automated penalty or promotional discount system.	Informs the highly modular design of the application to handle future expansion.
Ibrahim	2021	Hall allocation challenges in Nigeria	Documented administrative problems but did not propose a functional digital solution.	Corroborates the specific operational problem statement observed at LASU.
Kumar	2022	Review of event management technologies	Acted as a review of existing tools rather than proposing a custom institutional build.	Supports the choice of specific technologies like automated receipt generation.
Li	2017	Online booking system design	Proposed a generic booking logic that does not scale to complex university hierarchies.	Establishes the foundational logic required for reliable online reservation forms.
Lin	2023	Technology adoption in higher education	Focused on user behavior rather than algorithmic backend conflict resolution.	Reinforces the critical need for user training and an intuitive user interface.

Author(s)	Year	Core Focus	Identified Gap	Contribution to Current Project
Oyewale	2018	Facility management in Nigerian universities	Identified operational flaws across universities without building specialized software.	Provides localized empirical justification for immediate digital intervention.
Prashant Nikhare et al.	2025	Generic event management frameworks	Admitted that generic frameworks often fail to map to specific institutional rules.	Highlights the limitations of generic systems and the need for LASU-specific customization.
Thankachan et al.	2025	Multi-level approval architectures	Their approval chain did not include real-time financial gateway integration.	Informs the architectural design of the administrative dashboard and routing logic.
Turcu et al.	2015	ICT and RFID integration in campus life	Focused heavily on physical hardware integration rather than the software booking engine.	Provides the theoretical basis for future physical security enhancements.
Wiedyadari et al.	2024	Event analytics and reporting tools	Analytics focused primarily on attendance rather than financial revenue and damage penalties.	Inspires the integration of the comprehensive administrative analytics dashboard.
Xuan	2021	Secure room booking system implementation	Designed for library study rooms, lacking the complexity required for major event halls.	Emphasizes the importance of strict role-based access control and authentication.

Most existing systems provide generic booking features but lack customization for Nigerian university workflows and LASU's approval structure, which this project aims to address.

2.7 Summary of Literature Review

The review of conceptual, theoretical, and empirical literature unequivocally supports the transition from manual facility scheduling to automated, web based management systems. Studies consistently prove that digitalization reduces scheduling conflicts, accelerates administrative communication, and vastly improves institutional record keeping.

However, the literature also reveals that generic software solutions are inadequate for complex academic bureaucracies. To be truly effective, a system must natively understand the specific approval hierarchies and operational policies of the institution it serves. Furthermore, existing research predominantly ignores the post event lifecycle, treating facility management purely as a prevent scheduling problem. By addressing these gaps, developing a context specific, end to end digital workflow for Lagos State University represents both a practical operational necessity and a meaningful contribution to the field of educational technology

CHAPTER THREE

METHODOLOGY

3.1 System Overview

The methodology adopted for the Event Hall Management System (EVMS) revolves around digitizing complex institutional bureaucracy through a highly structured, state driven software architecture. At its core, the project applies object oriented programming principles and relational database design to solve the logistical challenges of facility reservation at Lagos State University.

Unlike conventional booking applications that rely on single tier approvals, this system was engineered from first principles to reflect the actual tripartite administrative structure of the university. It distributes authority across three distinct operational units: the Ventures Unit (handling commercial policy and final approvals), Facility Management (handling physical readiness and damage assessment), and the Bursary Unit (handling financial verification). By hardcoding this separation of concerns into the software workflow engine, the system eliminates the ambiguity and bottlenecks characteristic of manual paper pushing.

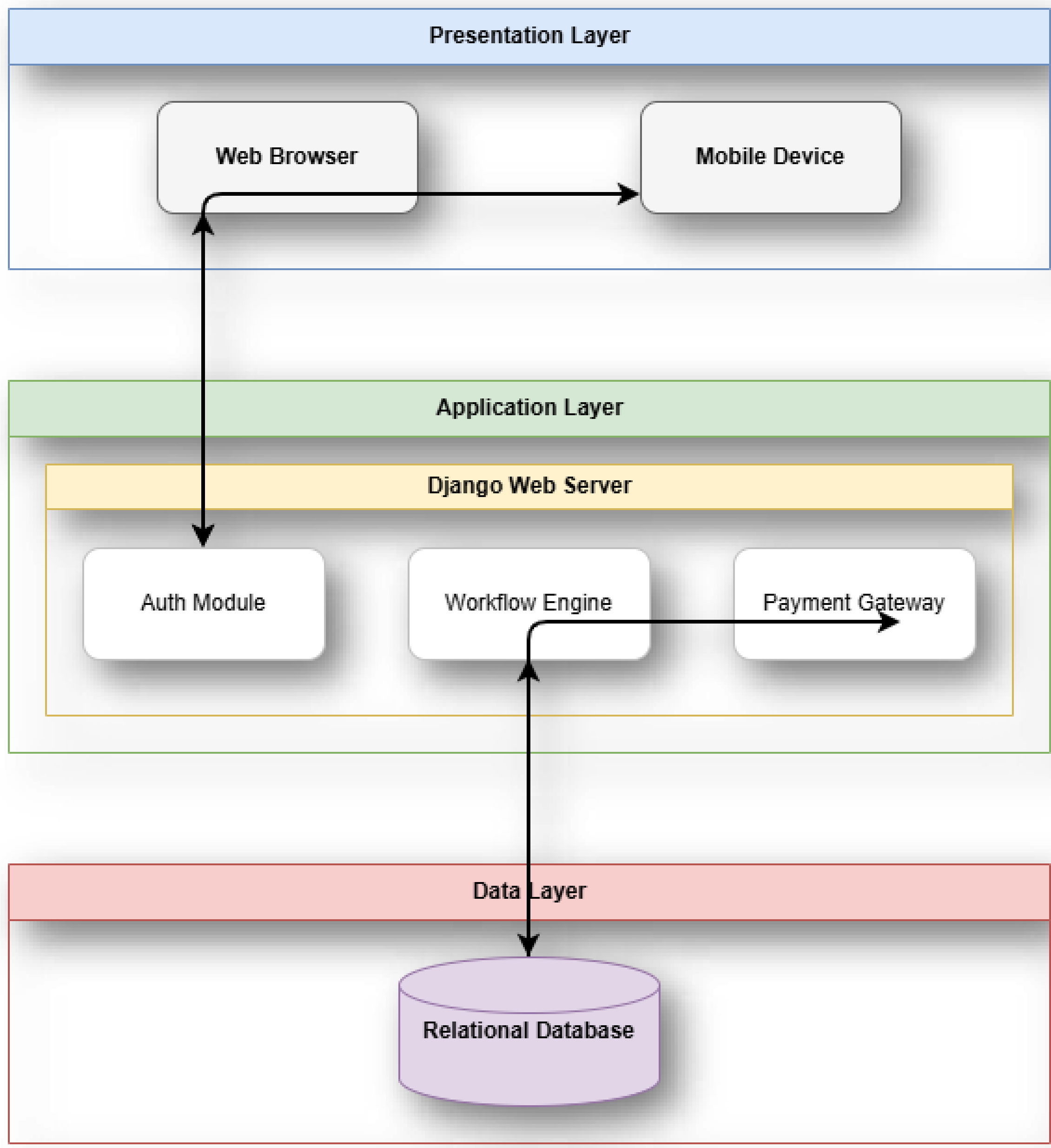


Figure 3.1 System Architecture

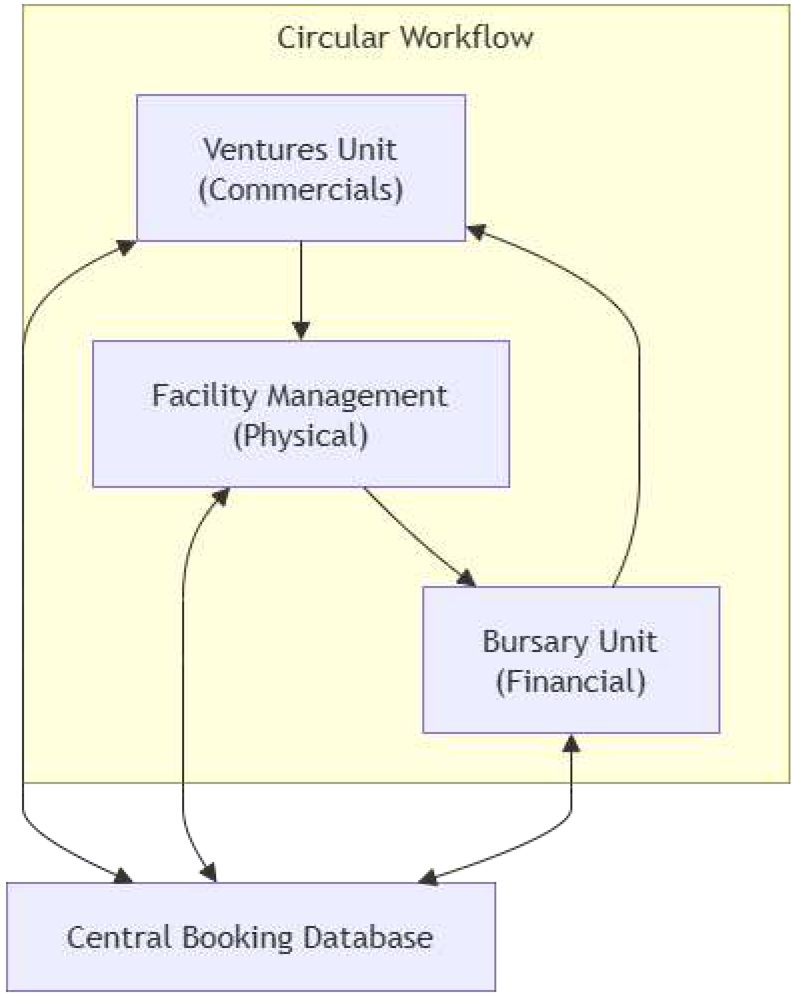
3.2 System Design and Components

The EVMS is constructed as a monolithic web application using the Django framework. The architecture is modularized into specific Django applications, ensuring that domain logic remains isolated and maintainable.

3.2.1 Tripartite Departmental Integration

The fundamental design decision of this project was to abandon a generic administrative backend in favor of unit specific dashboards and responsibilities.

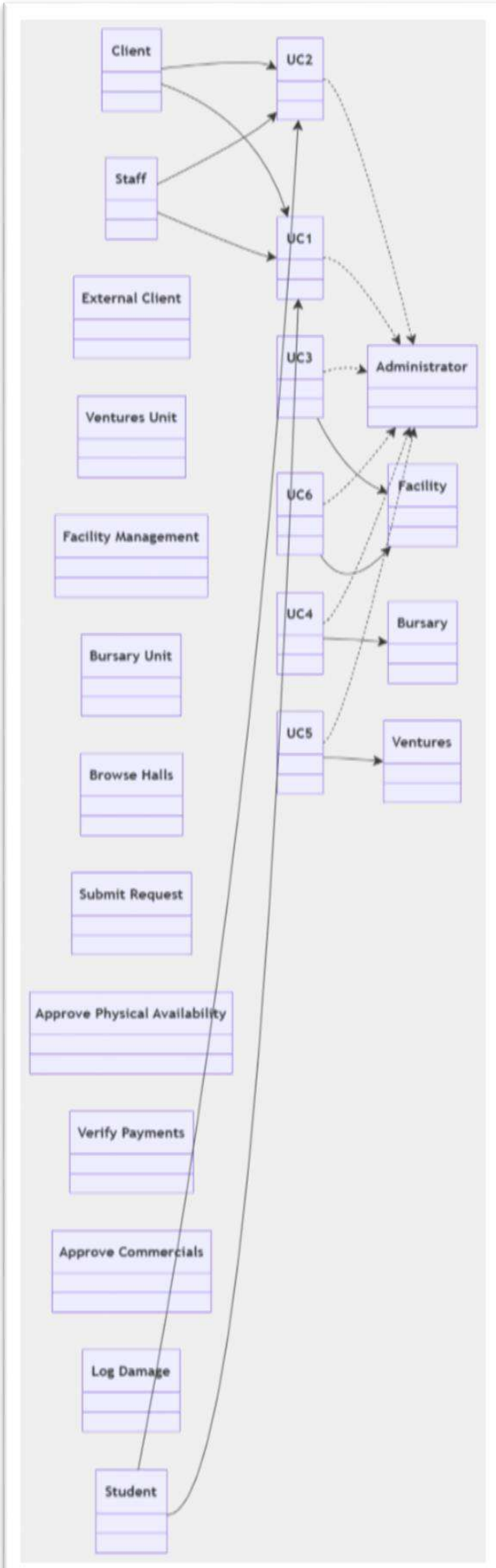
1. **Ventures Unit:** Acts as the primary commercial gatekeeper. They review initial applications, generate invoices, and issue the final confirmation only after both Facility and Bursary have completed their respective checks.
2. **Facility Management:** Acts as the physical gatekeeper. They possess the exclusive system capability to mark a hall as available or rejected based on maintenance schedules. Post event, they hold the authority to file official damage reports.
3. **Bursary Unit:** Acts as the financial gatekeeper. The inclusion of the Bursary Unit was a critical design iteration to ensure institutional financial compliance. Even when electronic payments are made, the Bursary Unit maintains a dedicated software queue (UNDER BURSARY VERIFICATION) where they verify payment proofs before the reservation can advance.



3.2.2 Role Based Access Control

Identity and access management are enforced using a custom user model and a dynamic capability system.

The system provisions seven primary roles:



Client
UC2
Staff
UC1
External Client
UC3
Administrator
Ventures Unit
Facility
UC6
Facility Management
UC4
Bursary
Bursary Unit
UC5
Ventures
Browse Halls
Submit Request
Approve Physical Availability
Verify Payments
Approve Commercials
Log Damage
Student

- **Student and External:** End users restricted to viewing their own applications and submitting payments.
- **Staff / Department:** University employees with booking privileges and limited access to departmental reports.
- **Ventures, Facility, and Bursary:** Administrative personnel with highly scoped permissions restricted to their specific operational domains.
- **Administrator:** Superusers with overarching system control.

Instead of hardcoding permissions into HTML templates, the system uses a backend service layer to evaluate if a user possesses the necessary capability before executing any state changing function. This design choice prevents privilege escalation vulnerabilities.

3.2.3 The State Machine Reservation Engine

The operational heart of the EVMS is its workflow engine. Facility booking is not treated as a simple data entry task; it is modeled as a strict state machine governed by a dedicated workflow service. A reservation must traverse a predefined sequence of statuses:

1. **Submission and Review:** The applicant creates a request. The Ventures Unit can place it under review or forward it to Facility Management.
2. **Physical Clearance:** Facility Management inspects the request and marks it available or rejected.
3. **Financial Clearance:** Ventures generates the invoice. The user uploads payment evidence or pays via gateway.
4. **Bursary Verification:** The system automatically routes the application to the Bursary Unit. The Bursary verifies the funds. If valid, the system advances. If invalid, Bursary can request clarification.

5. **Confirmation and Execution:** Ventures issues the final approval. On the event date, the status shifts to completed.
6. **Post Event Accountability:** Facility Management conducts an inspection. They either clear the event or log an incident, which triggers a secondary financial workflow requiring Bursary verification of the damage penalty payment.

This rigorous state management prevents illegal administrative actions, such as confirming an event before Bursary has verified the payment.

3.2.4 Coupon and Discount Component

To support flexible commercial operations, the system includes a dynamic discount engine. Administrators can generate promotional codes offering either percentage based or flat rate reductions.

The discount engine evaluates numerous constraints before applying a reduction. It verifies usage limits (both global and per user), date validity, minimum financial thresholds, and maximum discount caps. Furthermore, coupons can be hyper targeted, restricting their application to specific user roles, specific faculties, or specific categories of event halls. When a discount is successfully applied during the invoice generation stage, the system permanently records a snapshot of the coupon's value against the reservation, ensuring the financial audit trail remains intact even if the original coupon is later deleted.

3.2.5 Accountability and Penalty Component

Unlike many systems that ignore post event reality, the EVMS enforces strict accountability. The system features two punitive mechanisms.

First, administrators can issue manual penalties for policy violations, such as late cancellations. Second, Facility Management can log official damage reports linked to specific post event inspections.

Crucially, the system utilizes these records as blocking mechanisms. Any user account possessing an unpaid penalty or an unresolved damage report is programmatically prohibited from submitting new reservation requests. This feature actively protects university infrastructure and ensures prompt settlement of liabilities.

3.2.6 Notification System Component

To eliminate the communication delays inherent in manual processing, the system incorporates an aggressive notification strategy.

A central dispatcher generates alerts triggered by state changes in the workflow engine. For example, when Facility Management marks a hall as available, the Ventures Unit is instantly notified to generate the invoice. When payment succeeds, the applicant receives immediate confirmation.

These notifications are delivered via an internal web dashboard inbox and simultaneously dispatched as transactional emails via the Mailgun API. The system also supports administrative broadcast messaging, allowing management to distribute policy updates to targeted user segments, such as all registered students.

3.2.7 Payment and Transaction Component

Financial processing is fully integrated using the Paystack payment gateway, chosen for its reliability and dominance in the Nigerian digital economy.

When a user initiates payment, the system communicates with Paystack to generate a secure transaction session. The user completes the payment on Paystack's hosted infrastructure, minimizing security liabilities for the university. Upon completion, Paystack silently communicates back to the EVMS server via a secure webhook callback. The system independently verifies the transaction authenticity with Paystack's servers before automatically advancing the booking status and generating a downloadable PDF receipt.

For logistical flexibility, the system also provides a secure administrative interface for the Ventures Unit to manually record cash or direct bank transfer payments.

3.2.8 Reporting and Analytics Component

To support strategic facility management, the application features an analytical dashboard accessible to authorized personnel.

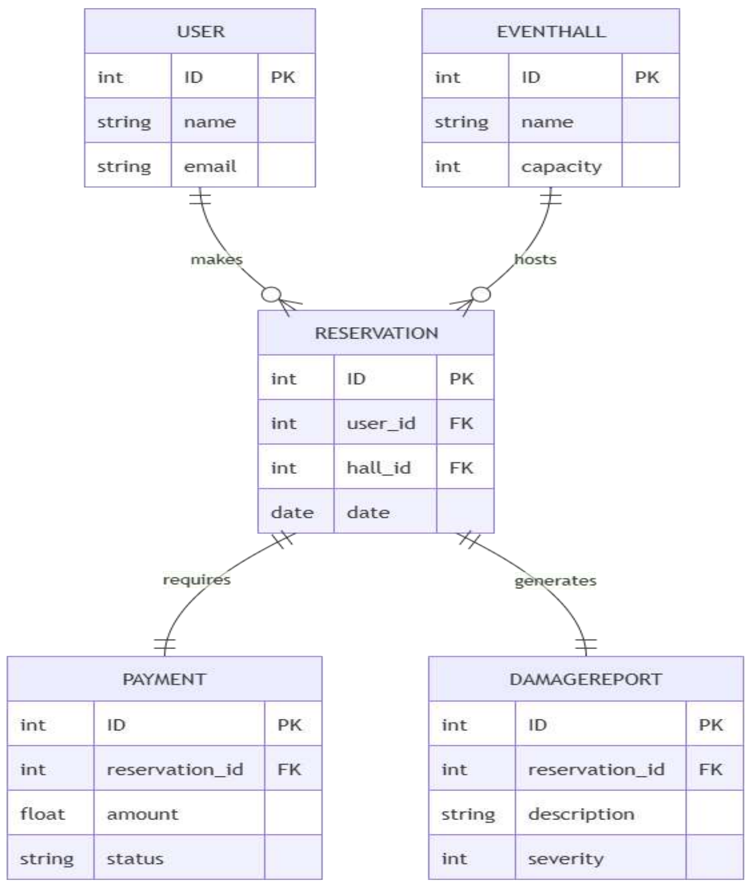
The reporting engine aggregates live database metrics and visualizes them using interactive JavaScript charts. Administrators can instantly view daily, weekly, or monthly booking volumes. They can analyze revenue streams segmented by standard booking fees, damage reparations, and penalty collections. The system automatically identifies the most frequently utilized halls and pinpoints peak booking hours.

All aggregated data can be exported into CSV files for external analysis, XLSX spreadsheets for accounting purposes, and formatted PDF documents for formal institutional reporting.

3.3 Algorithmic Conflict Detection

To solve the persistent problem of double booking, the system relies on a mathematical validation layer executed at the database transaction level.

When a booking form is submitted, the reservation model executes a query against the central database. It retrieves all existing reservations for the target hall on the specified date that are not in a cancelled or rejected state. The algorithm then evaluates the requested start and end times against the existing time blocks. If any overlap is detected, a validation error is thrown before the database write operation can occur. This design guarantees perfect chronological integrity, shifting the burden of conflict checking from human administrators to the server CPU.



CHAPTER FOUR

IMPLEMENTATION AND TESTING

4.1 Introduction

This This chapter details the technical execution of the EVMS design. It outlines the specific technology stack utilized, explains the programmatic implementation of the workflow engine and security measures, and discusses the testing methodologies employed to validate the system resilience under practical operational scenarios.

4.2 Technology Stack

The development environment utilized modern, industry standard web technologies to ensure a scalable and secure application.

- **Backend Framework (Django 6.0.4 with Python 3.11):** Selected for its comprehensive toolset. Django provided the necessary Object Relational Mapping tools, robust session security, and a highly customizable administration panel(Foundation, 2024).
- **Database Engine (SQLite):** Utilized for the development phase due to its zero configuration requirements, with the ORM models fully structured to support a seamless migration to PostgreSQL in a production environment.(Hipp, 2022; Owens, 2006)
- **Frontend Presentation (Bootstrap 5):** Employed to rapidly construct a mobile responsive, accessible user interface without the technical debt of maintaining a massive custom CSS codebase(Thornton, 2021).
- **Financial Integration (Paystack):** Integrated to handle secure, electronic transactions, abstracting the complexity of credit card processing away from the university servers (Paystack, 2024).

- **Document and Report Generation (ReportLab and OpenPyXL):** Utilized to programmatically render PDF receipts and compile complex Excel spreadsheets for administrative analytics(Lab, 2025).

4.3 Core System Implementations

4.3.1 Enforcing Atomic Workflow Transitions

The most complex engineering challenge was ensuring that the database never entered an inconsistent state during a workflow transition. For example, if the Bursary unit verifies a payment, the system must update the reservation status, create an immutable history record, log the action, and dispatch an email notification.

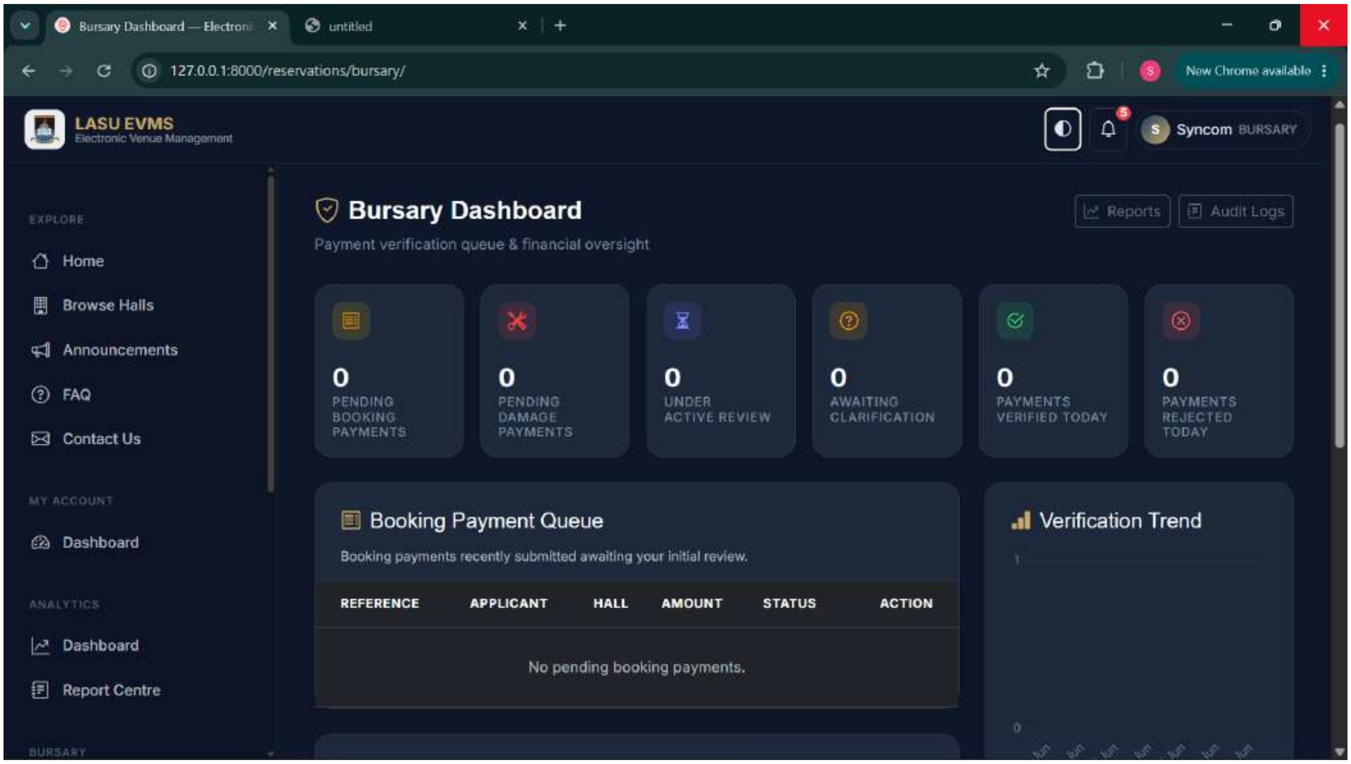
To achieve this reliably, all transitions were encapsulated within the workflow class using the database atomic context manager. This implementation guarantees that if the email dispatch fails due to a network timeout, the entire transaction rolls back. The database status will revert, and the audit logs will be deleted, ensuring that a partial state update never corrupts the application data.

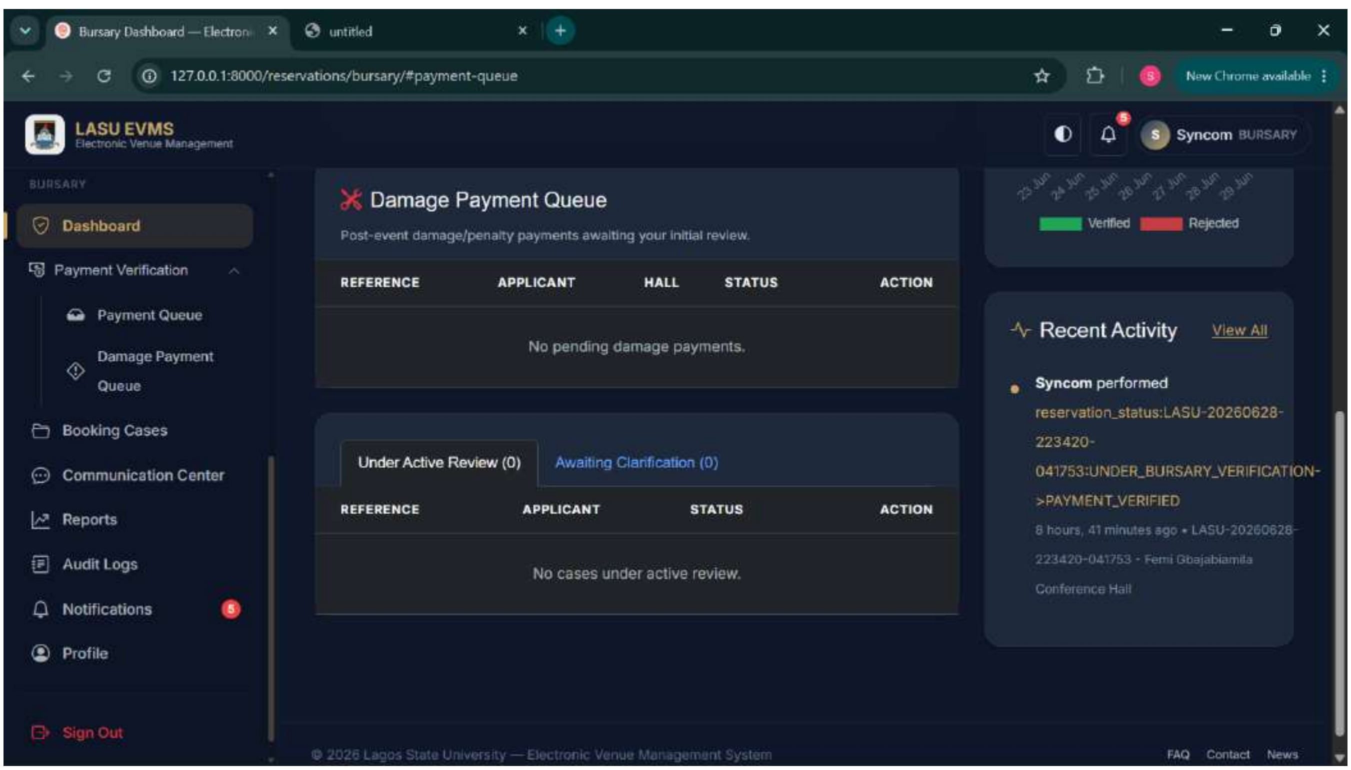
4.3.2 Implementation of the Bursary Verification Queue

Amazing payment experiences with the Paystack API (Paystack, 2024).

The inclusion of the Bursary Unit required specialized queue management logic. Functions such as verifying payments and rejecting payments were implemented.

When an applicant submits payment evidence, the system dynamically generates an internal notification targeted specifically at users possessing the Bursary role. The Bursary dashboard queries the database for all reservations currently awaiting verification. This targeted querying ensures that Bursary personnel are not overwhelmed with irrelevant data; they only see applications that explicitly require financial clearance.





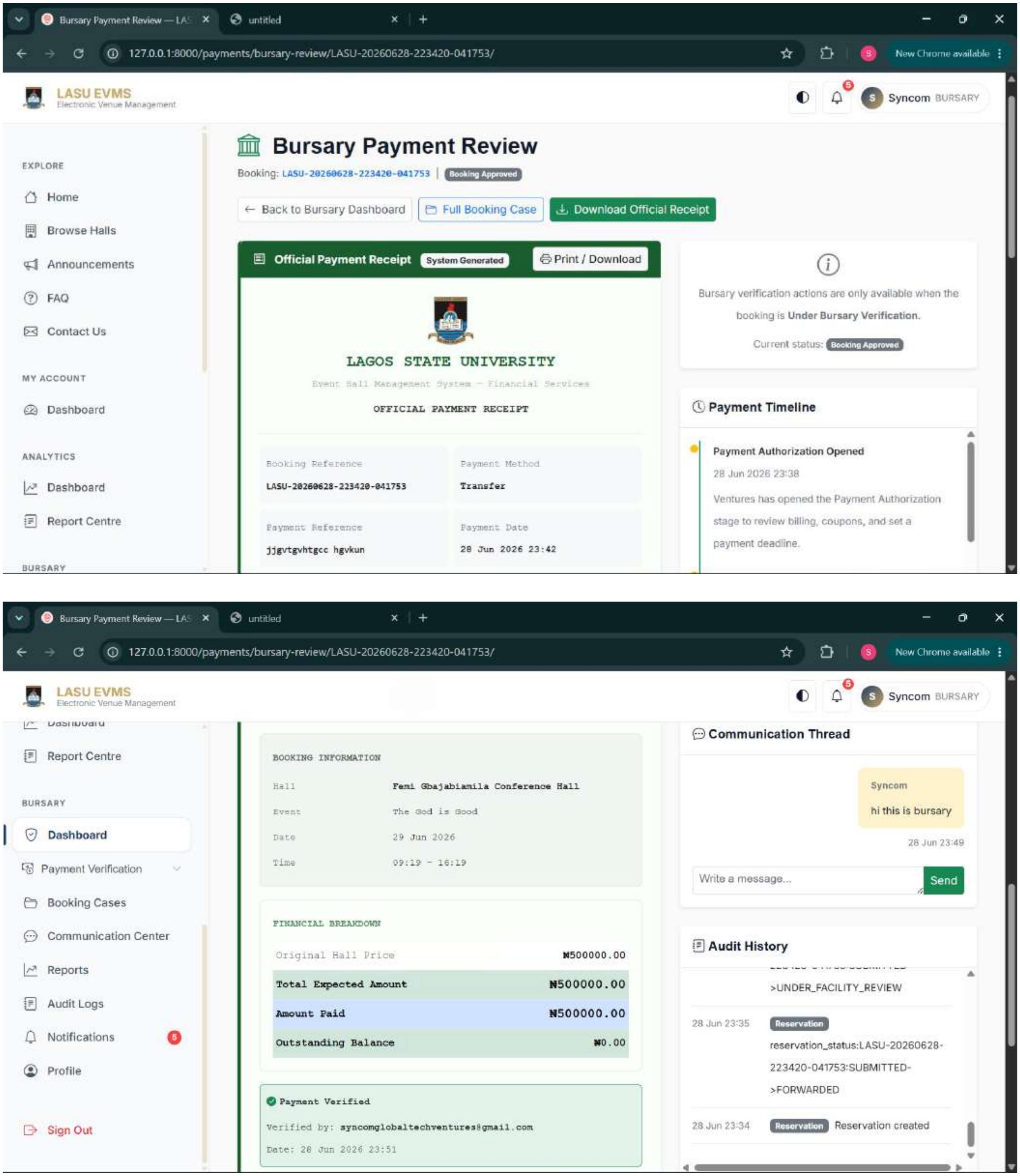
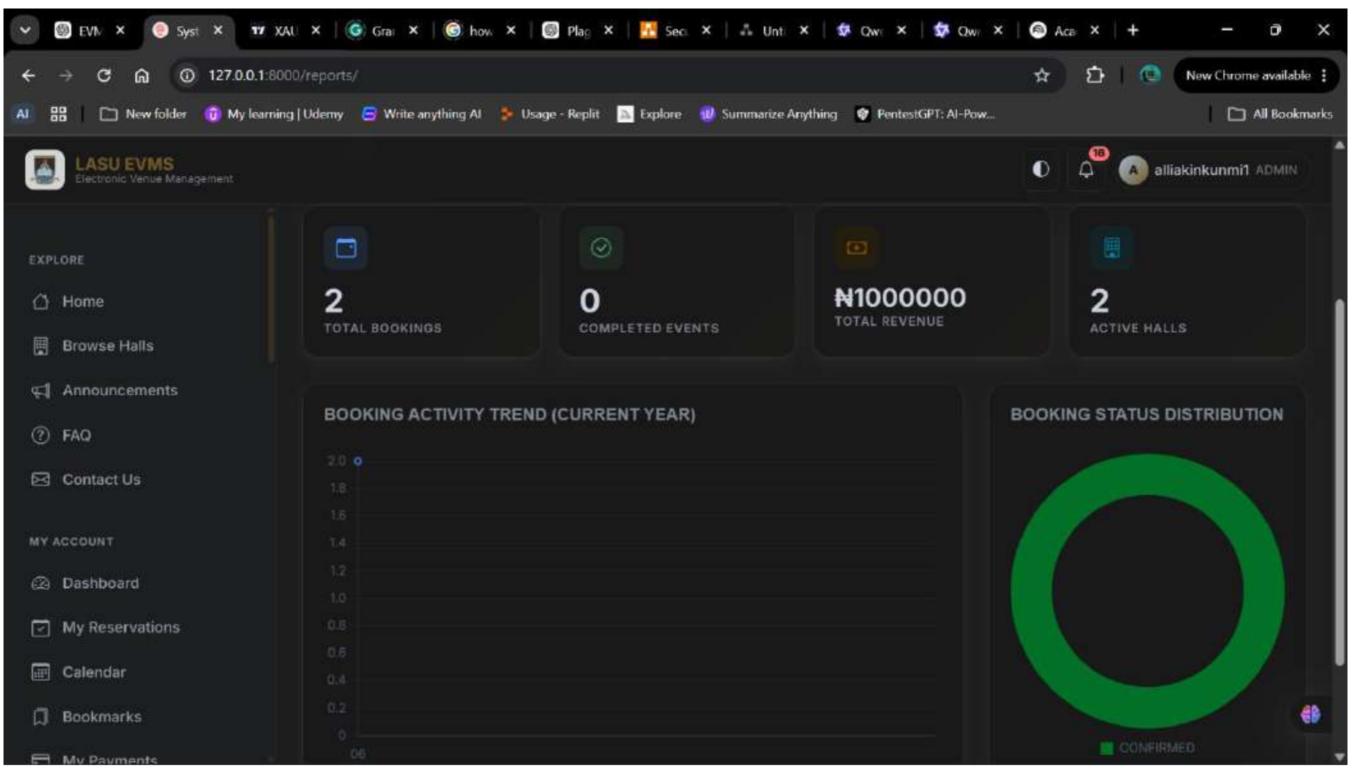


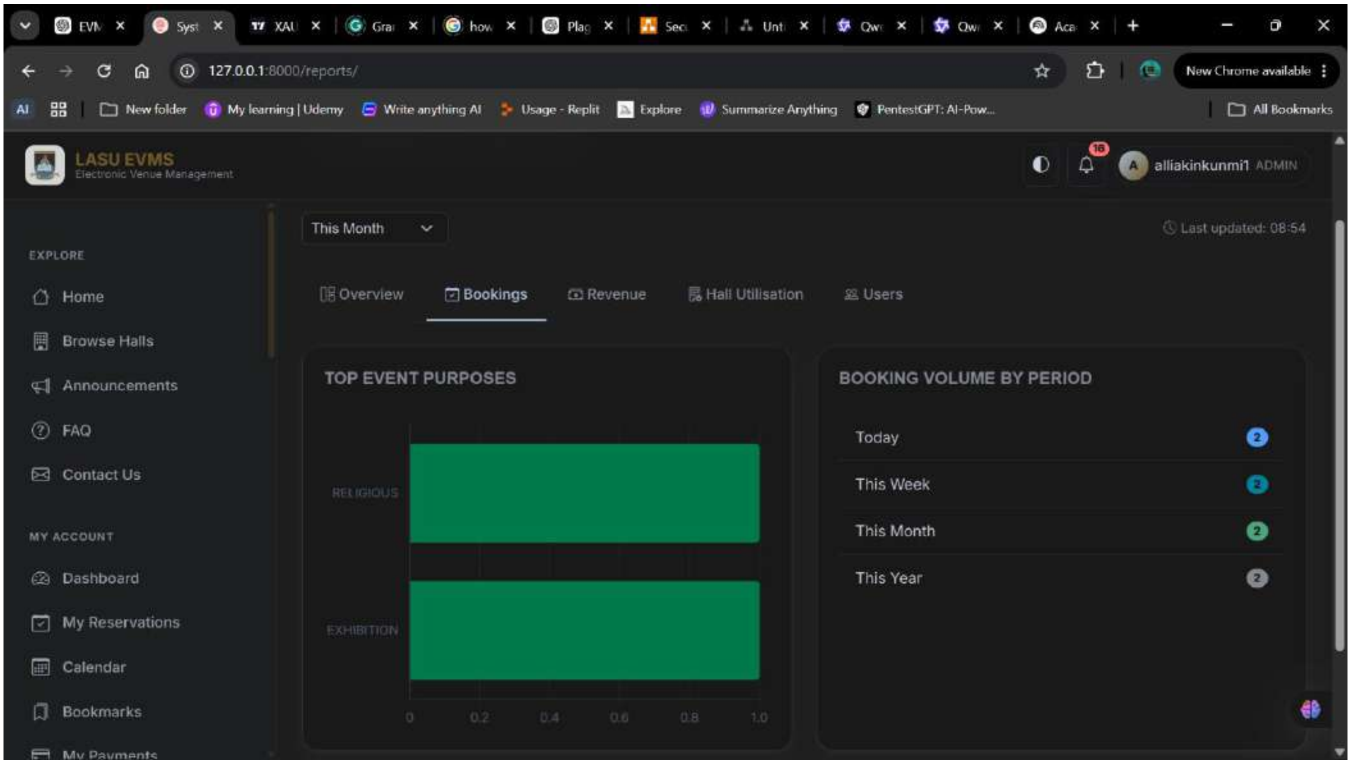
Figure 4.1: The Bursary Unit dashboard displaying the payment verification queue.

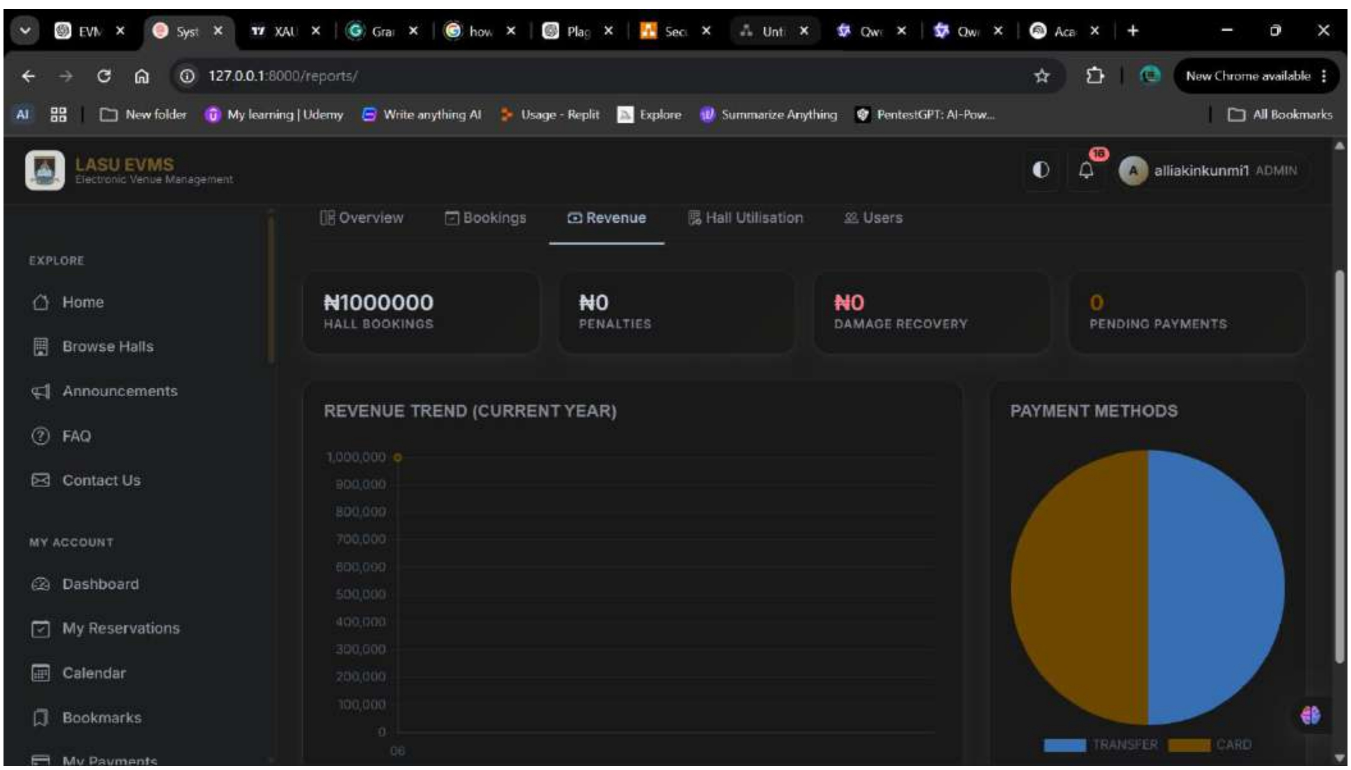
4.3.3 Dynamic Analytics and Dashboard Segregation

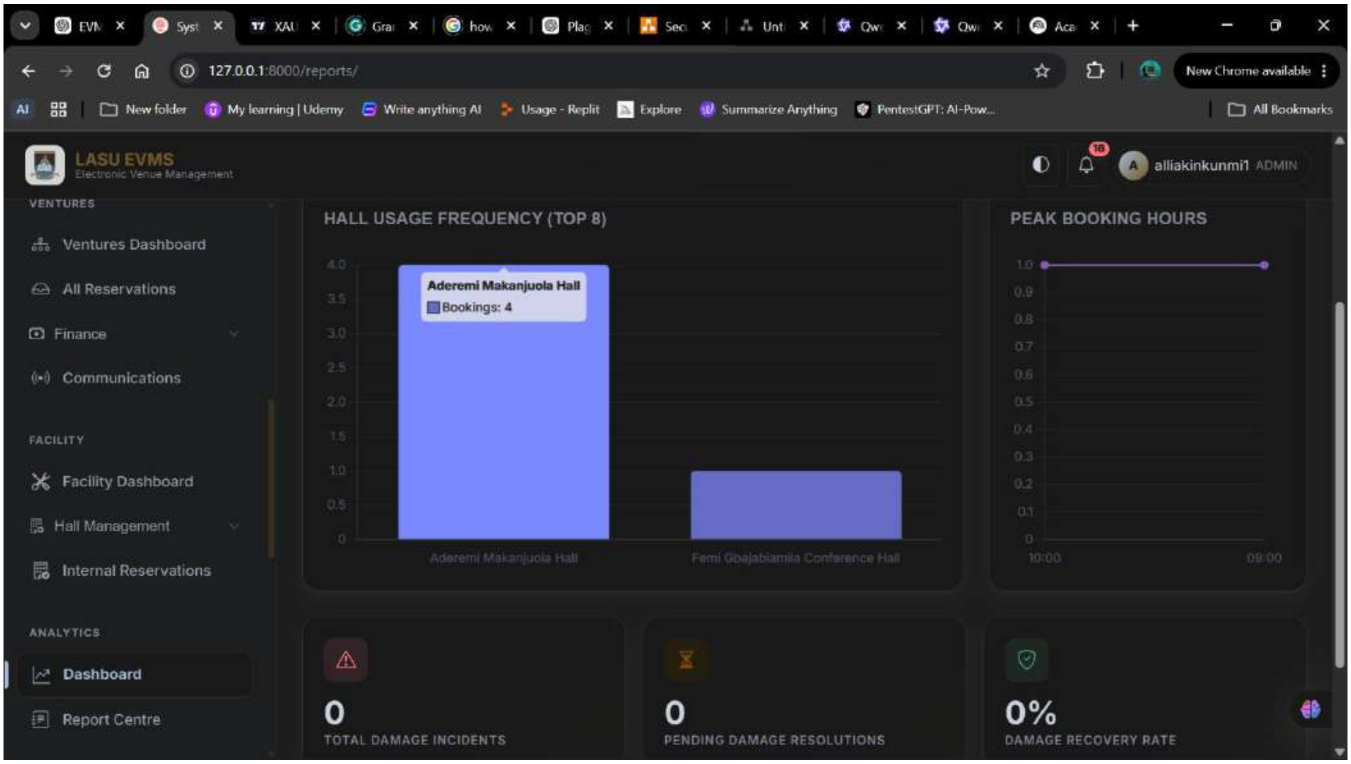
To fulfill the objective of providing actionable institutional data, a reporting engine was implemented. The system aggregates live database queries to calculate total revenue, segmenting it by standard booking fees, damage penalties, and late charges.

The presentation of this data is strictly governed by the user role. A dedicated view routing mechanism inspects the incoming request profile. If a student authenticates, they are directed to the standard dashboard. If a financial officer authenticates, the view intercepts the request and redirects them to the Bursary dashboard. This server side segregation ensures that unauthorized users cannot access sensitive administrative tools by simply manipulating the URL structure.

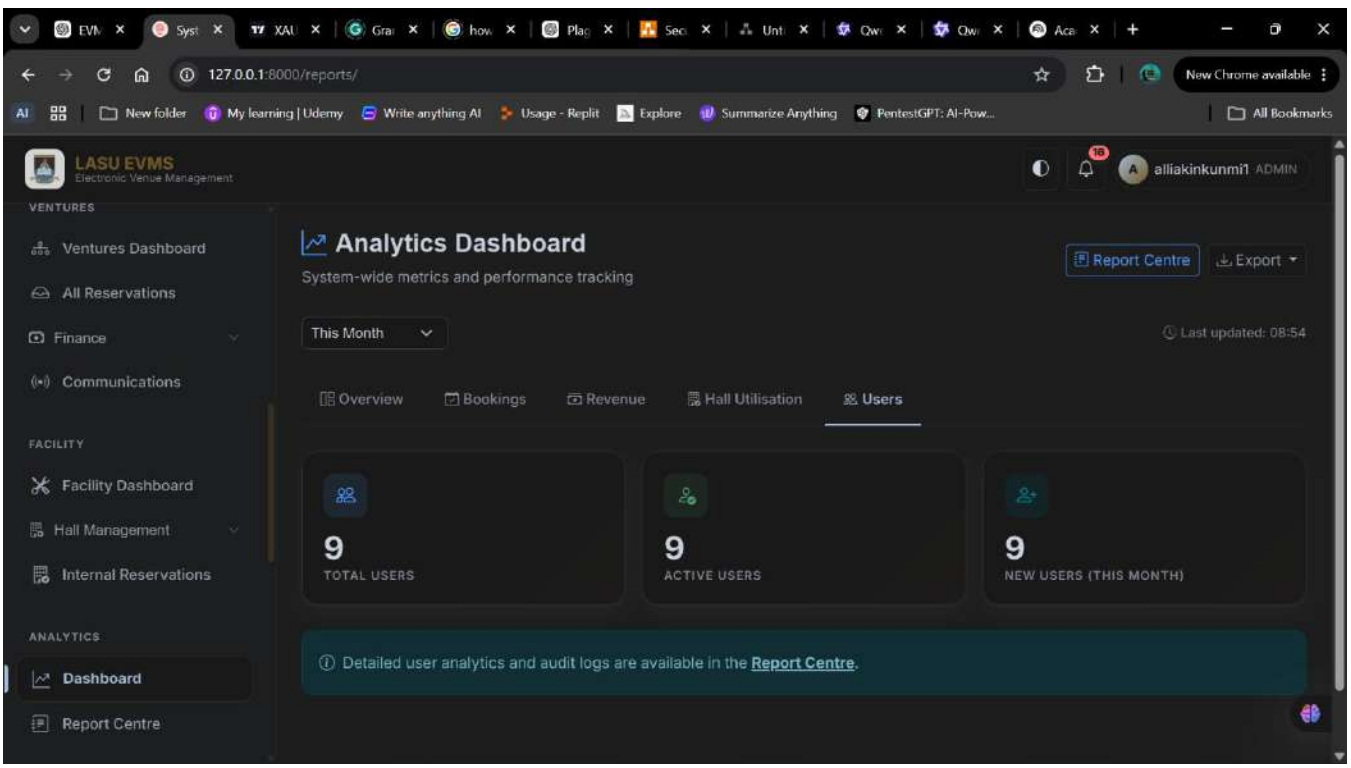



127.0.0.1:8000/reports/
LASU EVMS
Electronic Venue Management
alliakinkunmi1 ADMIN
This Month
Last updated: 08:54
EXPLORE
Home
Browse Halls
Announcements
FAQ
Contact Us
MY ACCOUNT
Dashboard
My Reservations
Calendar
Bookmarks
My Payments
Overview
Bookings
Revenue
Hall Utilisation
Users
TOP EVENT PURPOSES
RELIGIOUS
EXHIBITION
0
0.2
0.4
0.6
0.8
1.0
BOOKING VOLUME BY PERIOD
Today 2
This Week 2
This Month 2
This Year 2

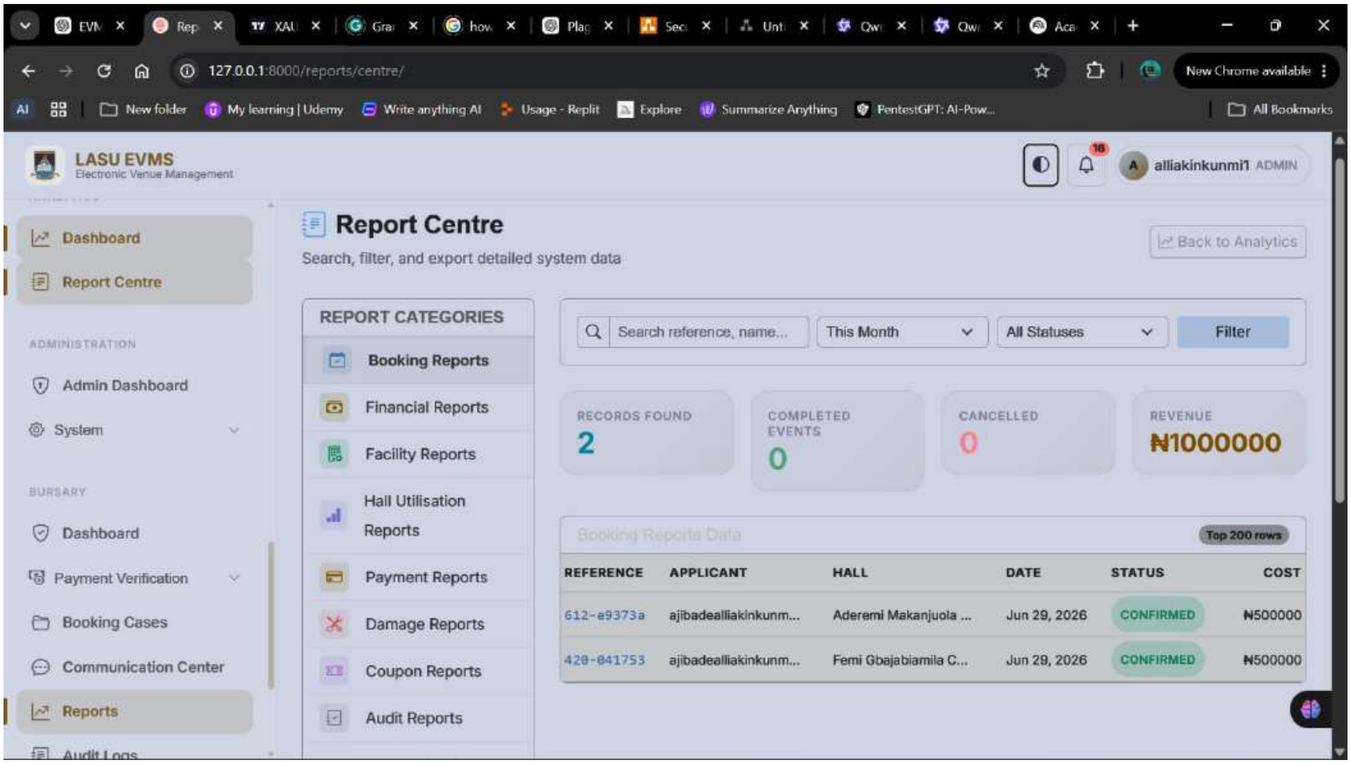

127.0.0.1:8000/reports/
LASU EVMS
Electronic Venue Management
alliakinkunmi1 ADMIN
EXPLORE
Home
Browse Halls
Announcements
FAQ
Contact Us
MY ACCOUNT
Dashboard
My Reservations
Calendar
Bookmarks
My Payments
Overview
Bookings
Revenue
Hall Utilisation
Users
₦1000000
HALL BOOKINGS
₦0
PENALTIES
₦0
DAMAGE RECOVERY
0
PENDING PAYMENTS
REVENUE TREND (CURRENT YEAR)
1,000,000
900,000
800,000
700,000
600,000
500,000
400,000
300,000
200,000
100,000
0
06
PAYMENT METHODS
TRANSFER
CARD



127.0.0.1:8000/reports/
LASU EVMS
Electronic Venue Management
alliakinkunmi1 ADMIN
VENTURES
Ventures Dashboard
All Reservations
Finance
Communications
FACILITY
Facility Dashboard
Hall Management
Internal Reservations
ANALYTICS
Dashboard
Report Centre
HALL USAGE FREQUENCY (TOP 8)
Aderemi Makanjuola Hall
Bookings: 4
Aderemi Makanjuola Hall
Femi Gbajabiamila Conference Hall
PEAK BOOKING HOURS
10:00
09:00
0
TOTAL DAMAGE INCIDENTS
0
PENDING DAMAGE RESOLUTIONS
0%
DAMAGE RECOVERY RATE



127.0.0.1:8000/reports/
LASU EVMS
Electronic Venue Management
alliakinkunmi1 ADMIN
Analytics Dashboard
System-wide metrics and performance tracking
Report Centre
Export
This Month
Last updated: 08:54
Overview
Bookings
Revenue
Hall Utilisation
Users
9
TOTAL USERS
9
ACTIVE USERS
9
NEW USERS (THIS MONTH)
Detailed user analytics and audit logs are available in the Report Centre.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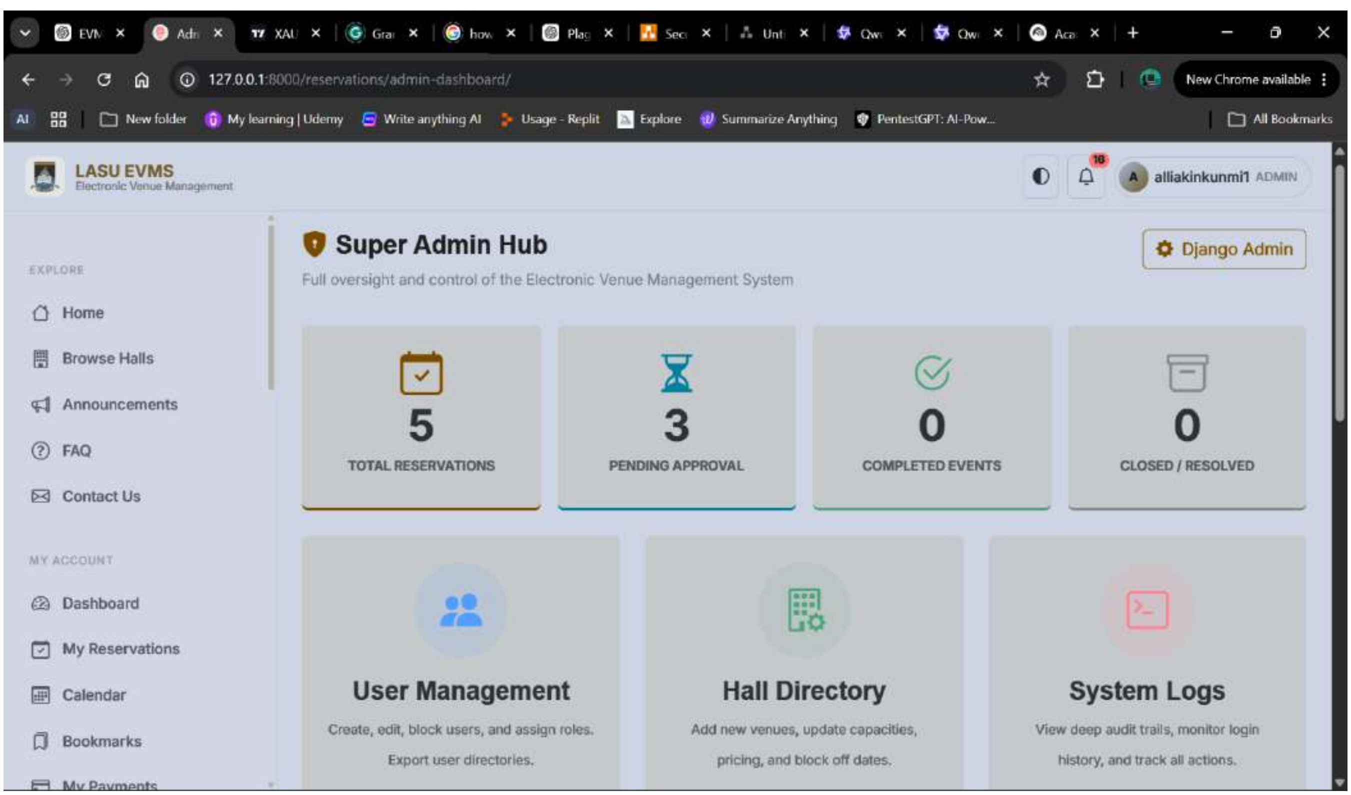
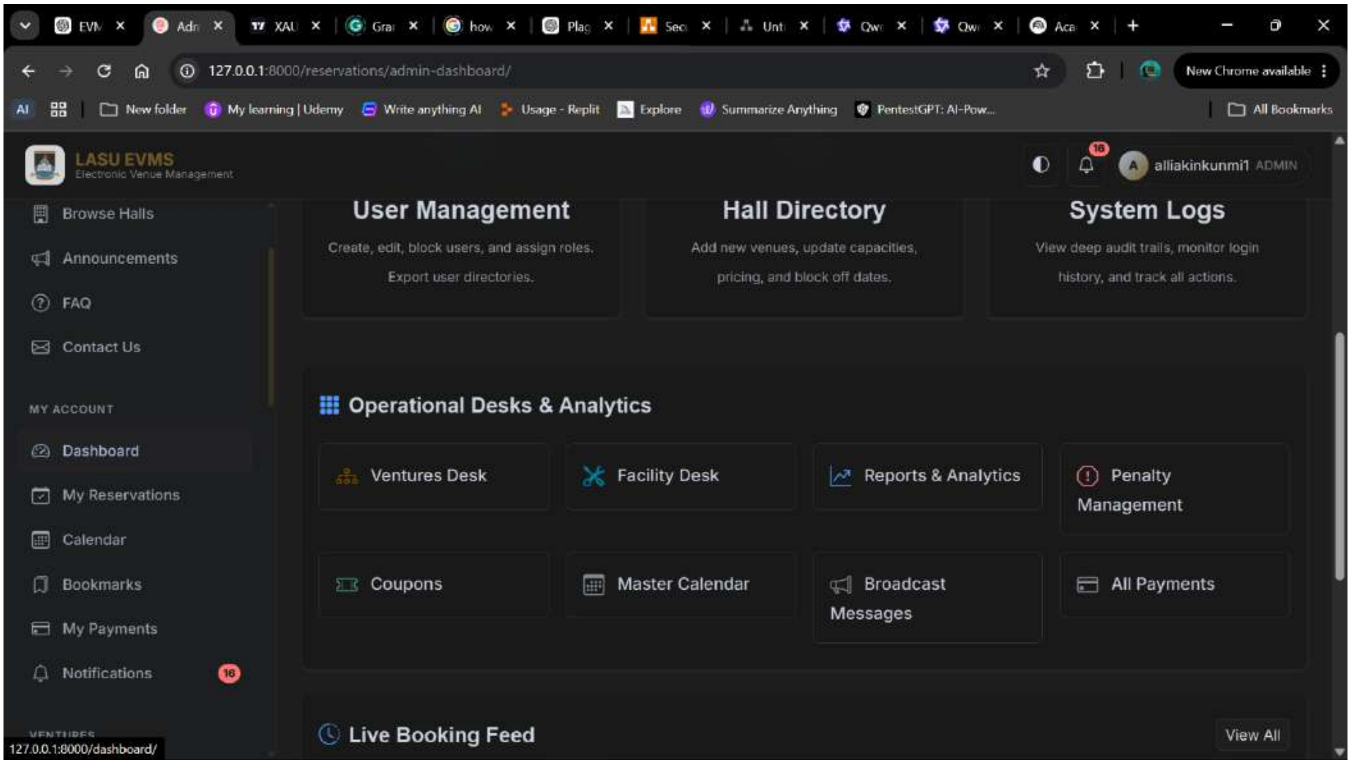
LASU EVMS
Electronic Venue Management
alliakinkunmi1 ADMIN
Dashboard
Report Centre
ADMINISTRATION
Admin Dashboard
System
BURSARY
Dashboard
Payment Verification
Booking Cases
Communication Center
Reports
Audit Logs
Report Centre
Search, filter, and export detailed system data
Back to Analytics
REPORT CATEGORIES
Booking Reports
Financial Reports
Facility Reports
Hall Utilisation Reports
Payment Reports
Damage Reports
Coupon Reports
Audit Reports
Search reference, name...
This Month
All Statuses
Filter
RECORDS FOUND 2
COMPLETED EVENTS 0
CANCELLED 0
REVENUE ₦1000000
Booking Reports Data
Top 200 rows
REFERENCE APPLICANT HALL DATE STATUS COST
612-e9373a ajibadealliakinkunm... Aderemi Makanjuola ... Jun 29, 2026 CONFIRMED ₦500000
420-041753 ajibadealliakinkunm... Femi Gbajabiamila C... Jun 29, 2026 CONFIRMED ₦500000


127.0.0.1:8000/reservations/admin-dashboard/
LASU EVMS
Electronic Venue Management
alliakinkunmi1 ADMIN
EXPLORE
Home
Browse Halls
Announcements
FAQ
Contact Us
MY ACCOUNT
Dashboard
My Reservations
Calendar
Bookmarks
My Payments
Super Admin Hub
Full oversight and control of the Electronic Venue Management System
Django Admin
5 TOTAL RESERVATIONS
3 PENDING APPROVAL
0 COMPLETED EVENTS
0 CLOSED / RESOLVED
User Management
Create, edit, block users, and assign roles. Export user directories.
Hall Directory
Add new venues, update capacities, pricing, and block off dates.
System Logs
View deep audit trails, monitor login history, and track all actions.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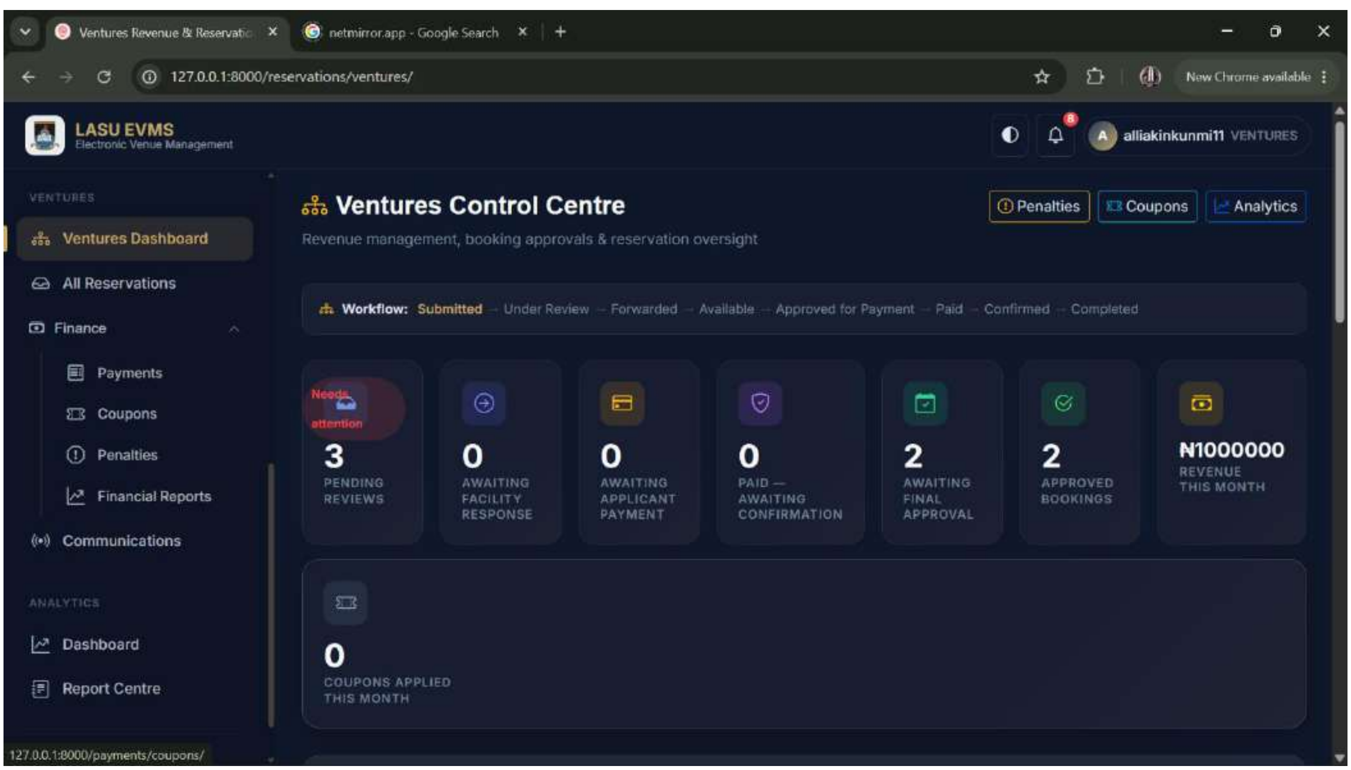
New Chrome available
New folder
My learning | Udemy
Write anything AI
Usage - Replit
Explore
Summarize Anything
PentestGPT: AI-Pow...
All Bookmarks
LASU EVMS
Electronic Venue Management
alliakinkunmi1 ADMIN
Browse Halls
Announcements
FAQ
Contact Us
MY ACCOUNT
Dashboard
My Reservations
Calendar
Bookmarks
My Payments
Notifications
User Management
Create, edit, block users, and assign roles. Export user directories.
Hall Directory
Add new venues, update capacities, pricing, and block off dates.
System Logs
View deep audit trails, monitor login history, and track all actions.
Operational Desks & Analytics
Ventures Desk
Facility Desk
Reports & Analytics
Penalty Management
Coupons
Master Calendar
Broadcast Messages
All Payments
Live Booking Feed
View All
127.0.0.1:8000/dashboard/


Ventures Revenue & Reservatio
netmirror.app - Google Search
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New Chrome available
LASU EVMS
Electronic Venue Management
alliakinkunmi11 VENTURES
VENTURES
Ventures Dashboard
All Reservations
Finance
Payments
Coupons
Penalties
Financial Reports
Communications
ANALYTICS
Dashboard
Report Centre
Ventures Control Centre
Revenue management, booking approvals & reservation oversight
Penalties
Coupons
Analytics
Workflow: Submitted → Under Review → Forwarded → Available → Approved for Payment → Paid → Confirmed → Completed
Needs attention
3
PENDING REVIEWS
0
AWAITING FACILITY RESPONSE
0
AWAITING APPLICANT PAYMENT
0
PAID — AWAITING CONFIRMATION
2
AWAITING FINAL APPROVAL
2
APPROVED BOOKINGS
₦1000000
REVENUE THIS MONTH
0
COUPONS APPLIED THIS MONTH
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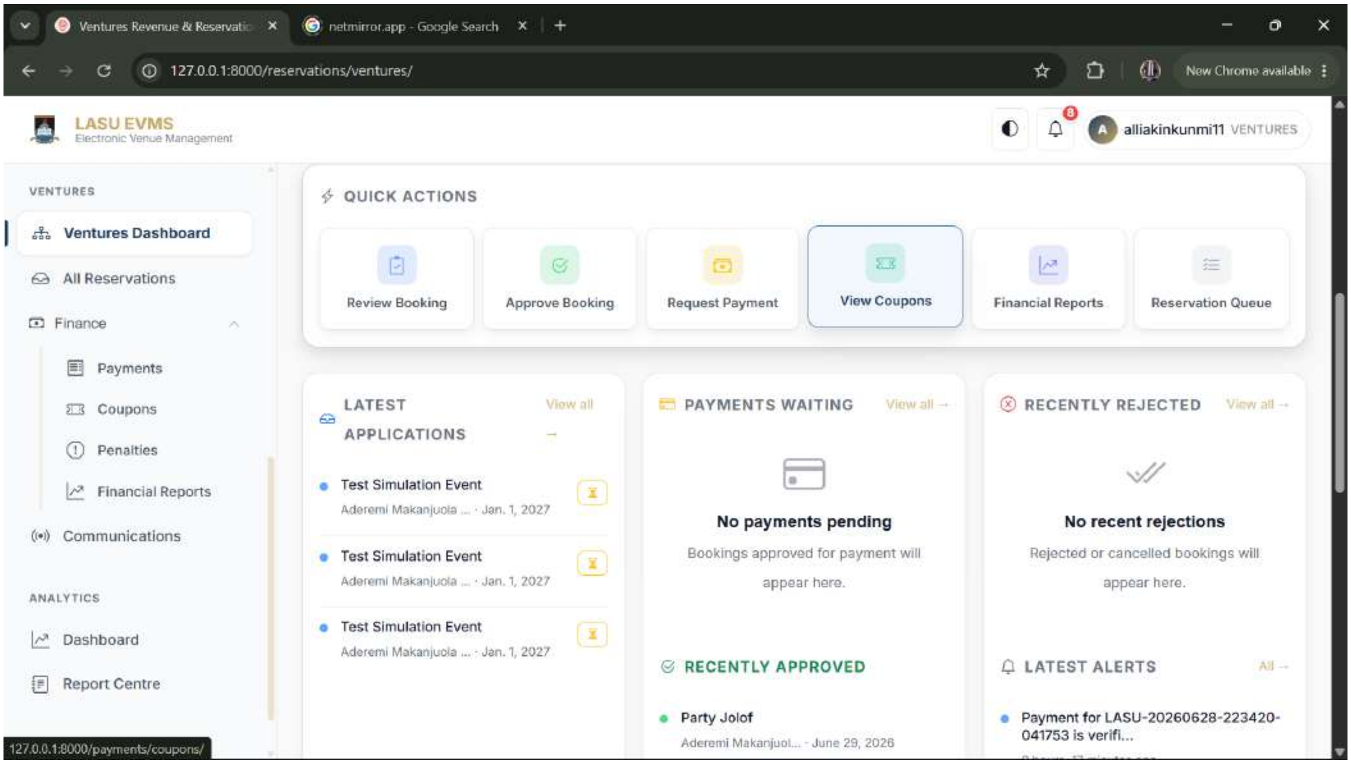
Ventures Revenue & Reservatio
netmirror.app - Google Search
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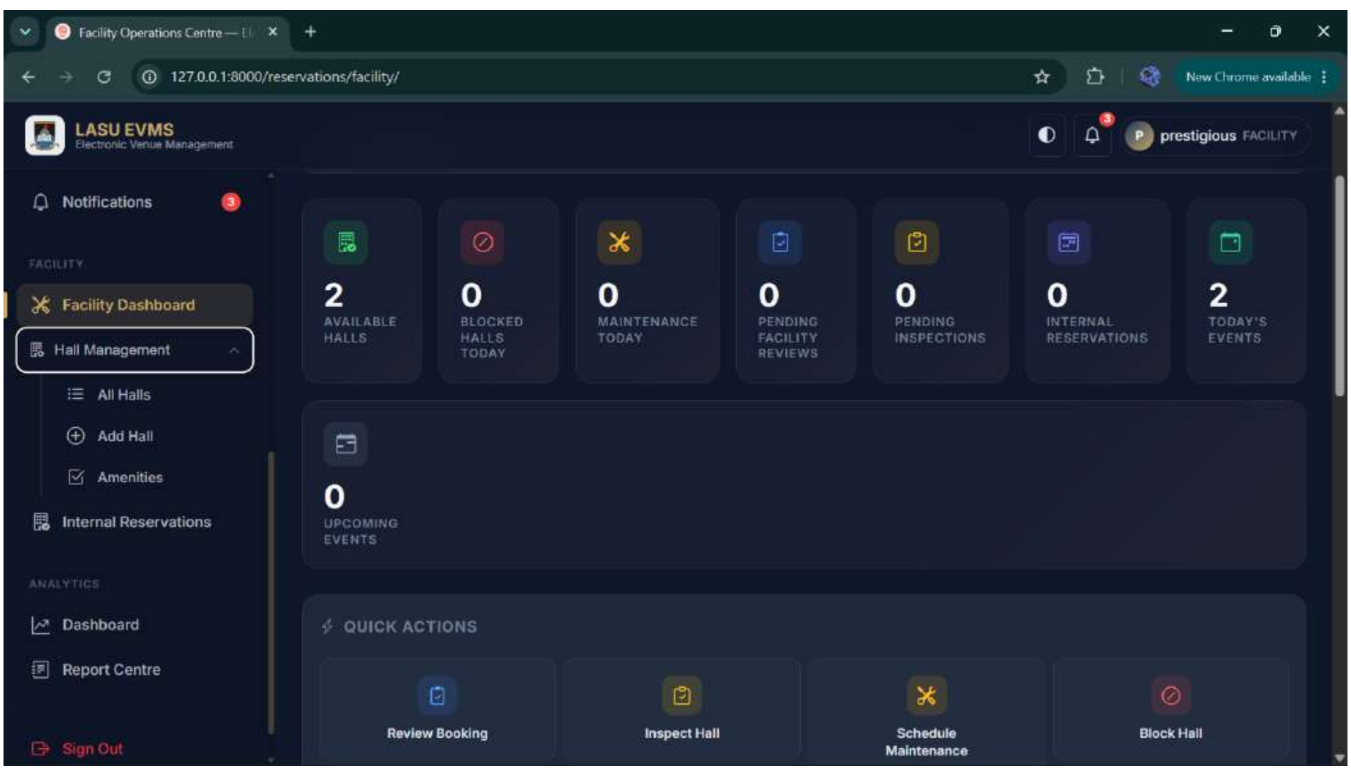
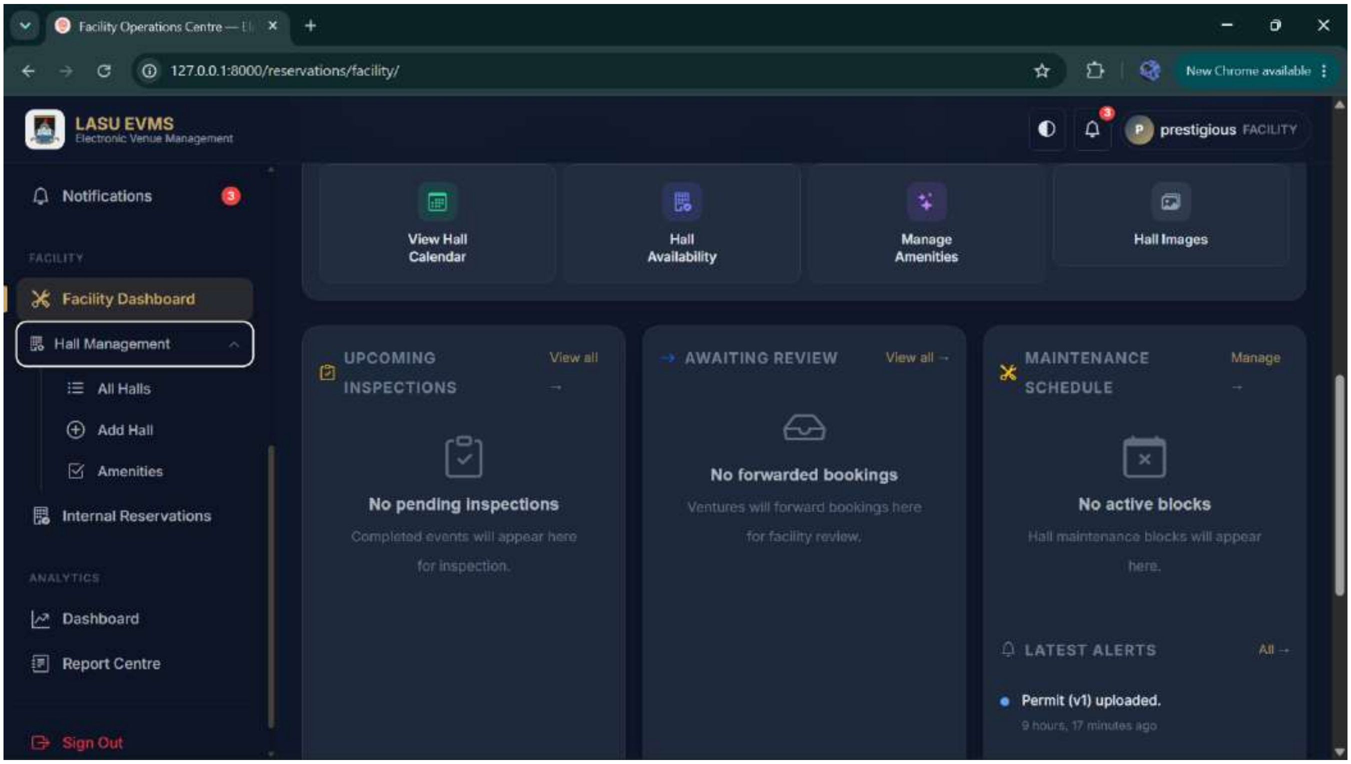
LASU EVMS
Electronic Venue Management
alliakinkunmi11 VENTURES
VENTURES
Ventures Dashboard
All Reservations
Finance
Payments
Coupons
Penalties
Financial Reports
Communications
ANALYTICS
Dashboard
Report Centre
QUICK ACTIONS
Review Booking
Approve Booking
Request Payment
View Coupons
Financial Reports
Reservation Queue
LATEST APPLICATIONS
View all
Test Simulation Event
Aderemi Makanjuola ... · Jan. 1, 2027
Test Simulation Event
Aderemi Makanjuola ... · Jan. 1, 2027
Test Simulation Event
Aderemi Makanjuola ... · Jan. 1, 2027
PAYMENTS WAITING
View all →
No payments pending
Bookings approved for payment will appear here.
RECENTLY APPROVED
Party Jolof
Aderemi Makanjuol... · June 29, 2026
RECENTLY REJECTED
View all →
No recent rejections
Rejected or cancelled bookings will appear here.
LATEST ALERTS
All →
Payment for LASU-20260628-223420-041753 is verifi...
127.0.0.1:8000/payments/coupons/


Facility Operations Centre — El
127.0.0.1:8000/reservations/facility/
New Chrome available
LASU EVMS
Electronic Venue Management
prestigious FACILITY
Notifications
FACILITY
Facility Dashboard
Hall Management
All Halls
Add Hall
Amenities
Internal Reservations
ANALYTICS
Dashboard
Report Centre
Sign Out
2
AVAILABLE HALLS
0
BLOCKED HALLS TODAY
0
MAINTENANCE TODAY
0
PENDING FACILITY REVIEWS
0
PENDING INSPECTIONS
0
INTERNAL RESERVATIONS
2
TODAY'S EVENTS
0
UPCOMING EVENTS
QUICK ACTIONS
Review Booking
Inspect Hall
Schedule Maintenance
Block Hall



Facility Operations Centre — El
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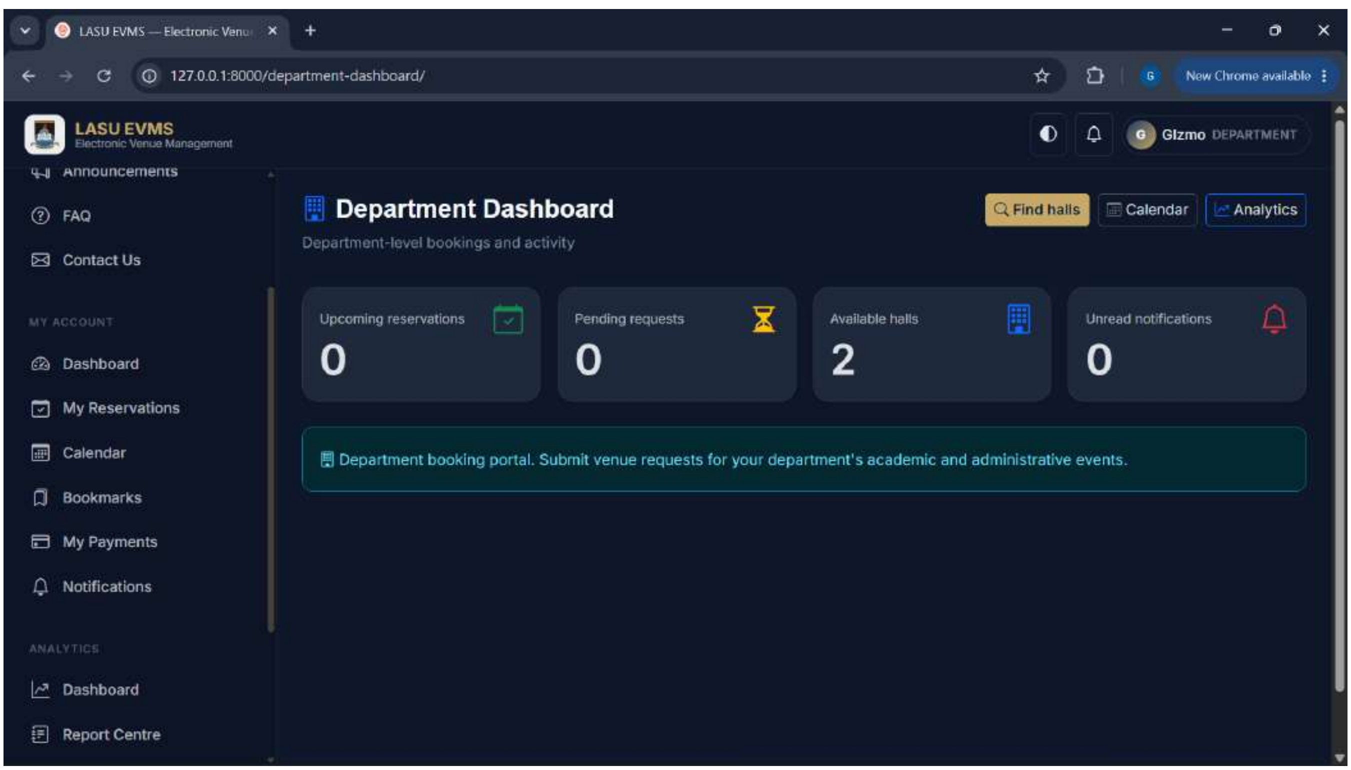
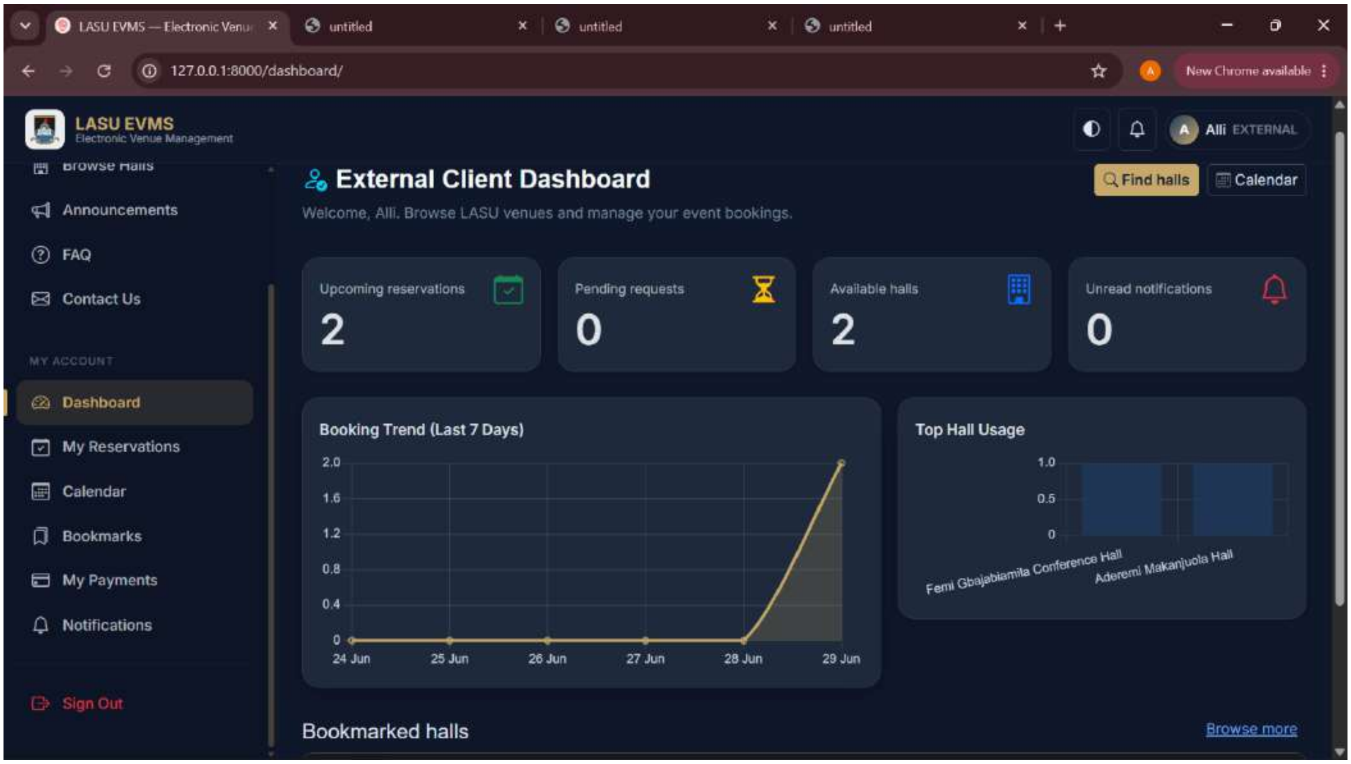
LASU EVMS
Electronic Venue Management
prestigious FACILITY
Notifications
FACILITY
Facility Dashboard
Hall Management
All Halls
Add Hall
Amenities
Internal Reservations
ANALYTICS
Dashboard
Report Centre
Sign Out
View Hall Calendar
Hall Availability
Manage Amenities
Hall Images
UPCOMING INSPECTIONS
View all →
No pending inspections
Completed events will appear here for inspection.
AWAITING REVIEW
View all →
No forwarded bookings
Ventures will forward bookings here for facility review.
MAINTENANCE SCHEDULE
Manage →
No active blocks
Hall maintenance blocks will appear here.
LATEST ALERTS
All →
Permit (v1) uploaded.
9 hours, 17 minutes ago



LASU EVMS — Electronic Venu
127.0.0.1:8000/department-dashboard/
New Chrome available
LASU EVMS
Electronic Venue Management
Gizmo DEPARTMENT
Announcements
FAQ
Contact Us
MY ACCOUNT
Dashboard
My Reservations
Calendar
Bookmarks
My Payments
Notifications
ANALYTICS
Dashboard
Report Centre
Department Dashboard
Department-level bookings and activity
Find halls
Calendar
Analytics
Upcoming reservations
0
Pending requests
0
Available halls
2
Unread notifications
0
Department booking portal. Submit venue requests for your department's academic and administrative events.



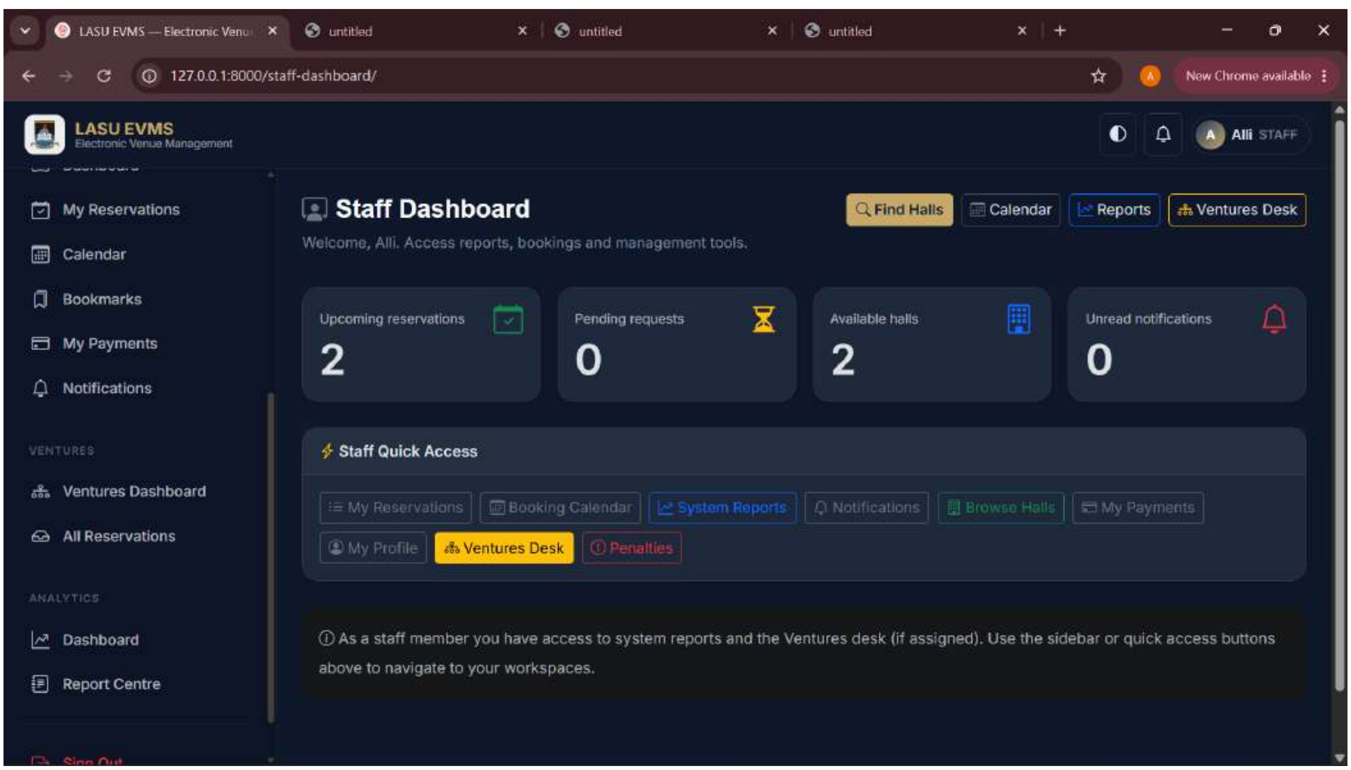


Figure 4.2: Dynamic administrative dashboard visualizing revenue and facility utilization metrics.

4.5 Testing and Validation

Testing was conducted iteratively throughout the development lifecycle to ensure functional correctness and security compliance.

4.5.1 Unit Testing of State Logic

Automated unit tests were written to target the workflow service. These tests simulated illegal administrative actions. For instance, a test script was designed to forcefully attempt a transition from submitted directly to paid, bypassing Facility Management and the Bursary Unit. The workflow service successfully intercepted the illegal status jump and raised a programmatic exception, validating the integrity of the state machine.

4.5.2 Functional Scenario Testing

Several practical scenarios were executed manually to observe system behavior.

Scenario 1: End to End Multidepartmental Approval

- **Execution:** A test student submitted a booking request. The Ventures account forwarded it. The Facility account marked it available. The Ventures account generated the invoice. The student uploaded a mock payment receipt. The Bursary account verified the receipt. Finally, the Ventures account confirmed the booking.
- **Outcome:** The system successfully transitioned through all requisite states. The audit trail accurately recorded the timestamps and actor identities at every stage, proving the viability of the tripartite approval architecture.

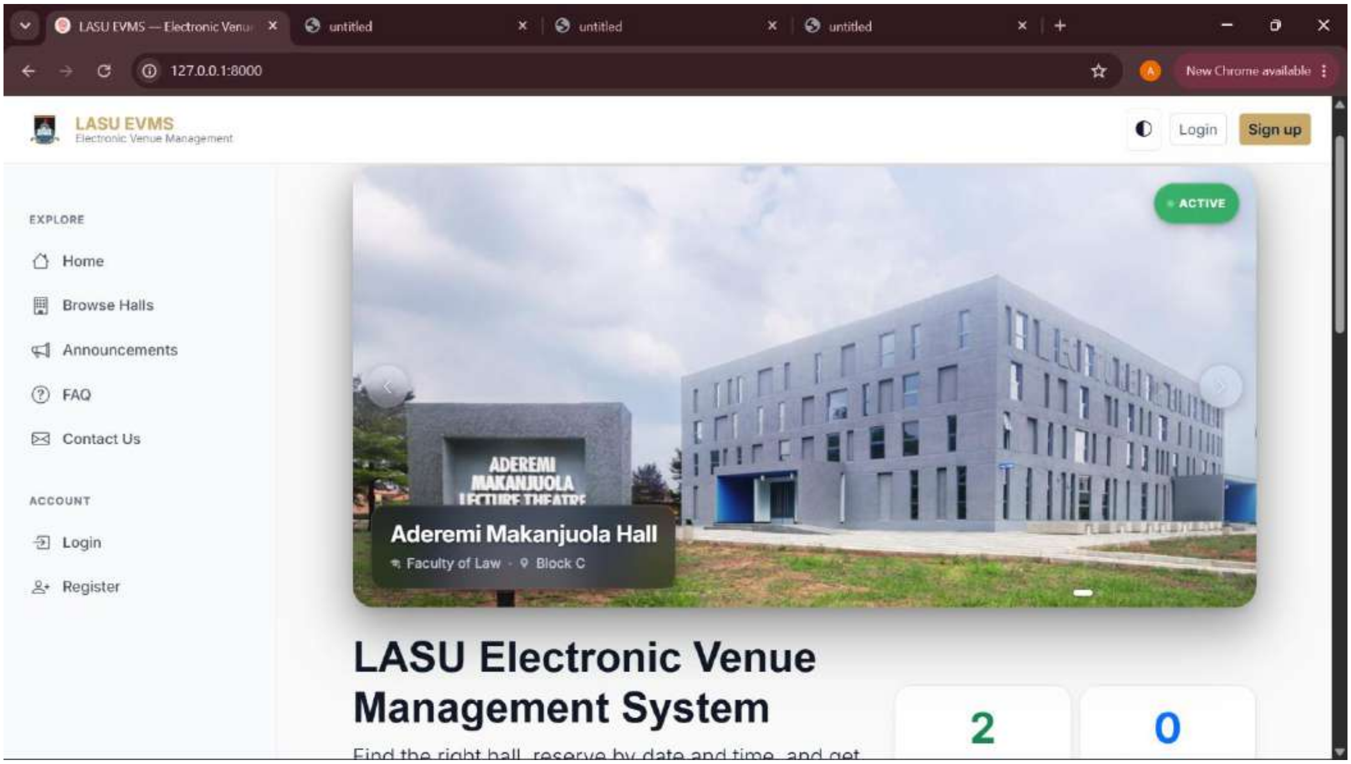
Scenario 2: Post Event Accountability Enforcement

- **Execution:** Facility Management logged a damage report against a completed event and assessed a penalty fee. The user account associated with the booking subsequently attempted to navigate to the new reservation form.
- **Outcome:** The system blocking middleware identified the unpaid penalty and successfully intercepted the request, redirecting the user to a mandatory payment page. This validated the ability of the system to automatically enforce institutional policy.

The implementation successfully addressed all objectives stated in Chapter One, including a 13-status multi-stage approval workflow unique to LASU's administrative structure, automated conflict detection, role-based access control for six distinct user types, Paystack-integrated payment processing, comprehensive analytics and multi-format report exports, and a full audit trail of all booking activities.

Testing results confirm that the system performs effectively, correctly, and securely. Identified limitations primarily relate to scalability concerns appropriate for a future production upgrade rather than functional deficiencies of the current implementation.

Overall, the system meets its design objectives and demonstrates the effectiveness of digital, workflow-driven solutions in improving university facility management (Chaturvedi et al., 2024; Thankachan et al., 2025; Wiedyadari et al., 2024).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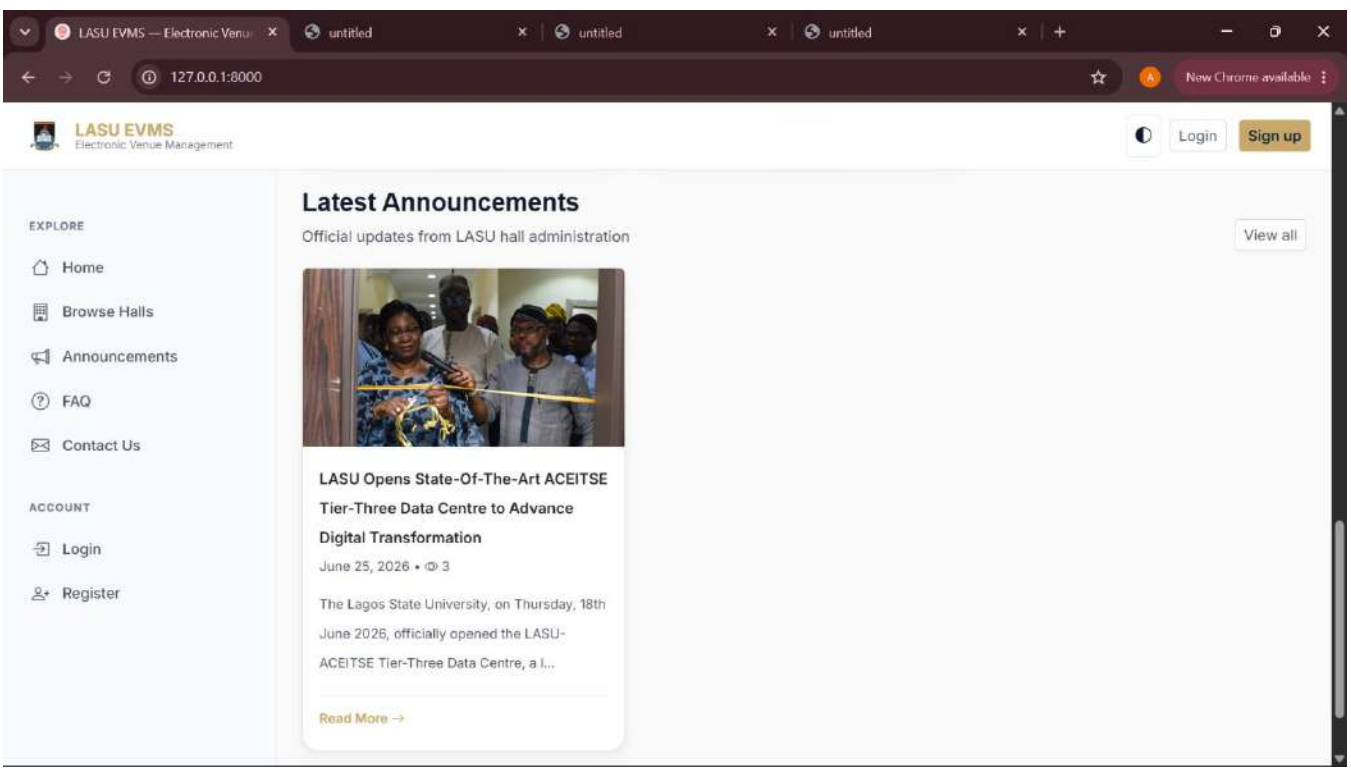
New Chrome available
LASU EVMS
Electronic Venue Management
Login
Sign up
EXPLORE
Home
Browse Halls
Announcements
FAQ
Contact Us
ACCOUNT
Login
Register
ACTIVE
ADEREMI
MAKANJUOLA
Aderemi Makanjuola Hall
Faculty of Law · Block C
LASU Electronic Venue
Management System
2
0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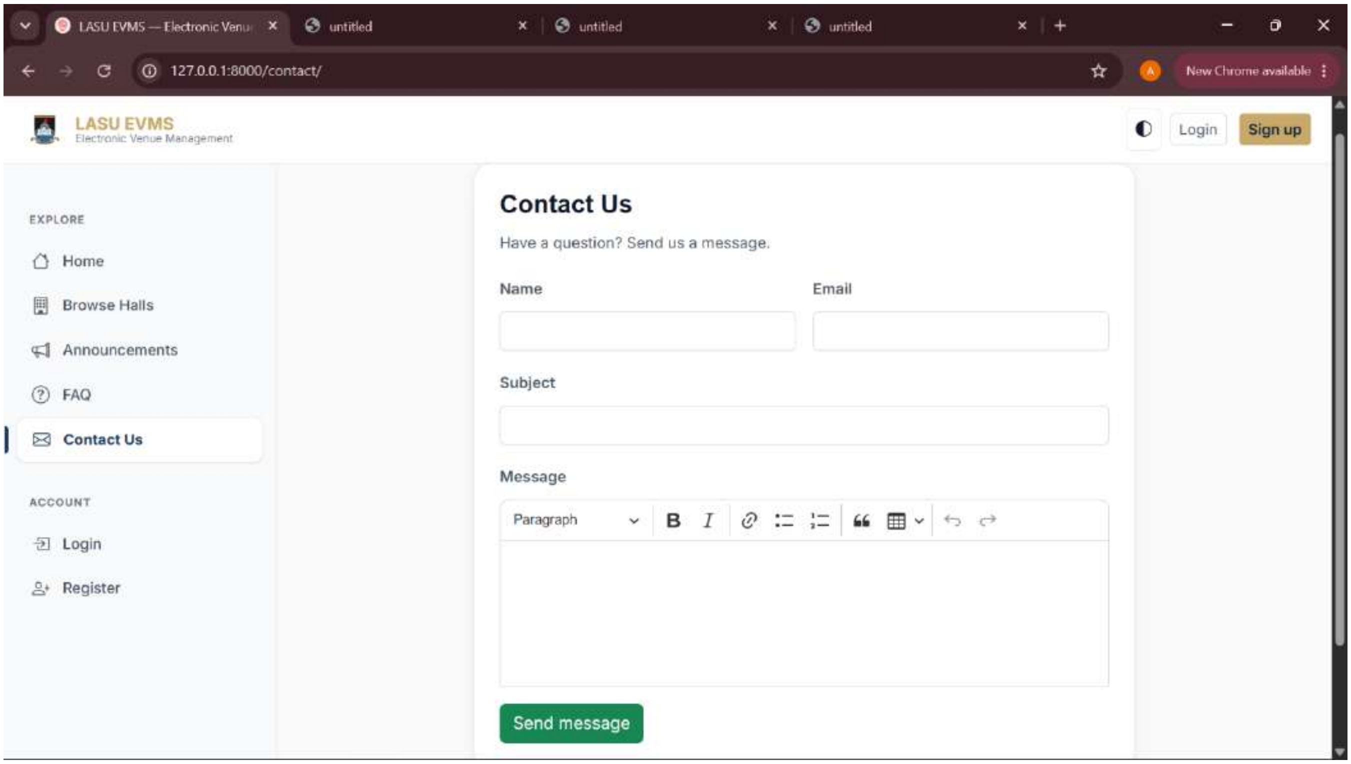
New Chrome available
LASU EVMS
Electronic Venue Management
Login
Sign up
EXPLORE
Home
Browse Halls
Announcements
FAQ
Contact Us
ACCOUNT
Login
Register
Latest Announcements
Official updates from LASU hall administration
View all
LASU Opens State-Of-The-Art ACEITSE Tier-Three Data Centre to Advance Digital Transformation
June 25, 2026 • 3
The Lagos State University, on Thursday, 18th June 2026, officially opened the LASU-ACEITSE Tier-Three Data Centre, a l...
Read More →


LASU EVMS — Electronic Venue
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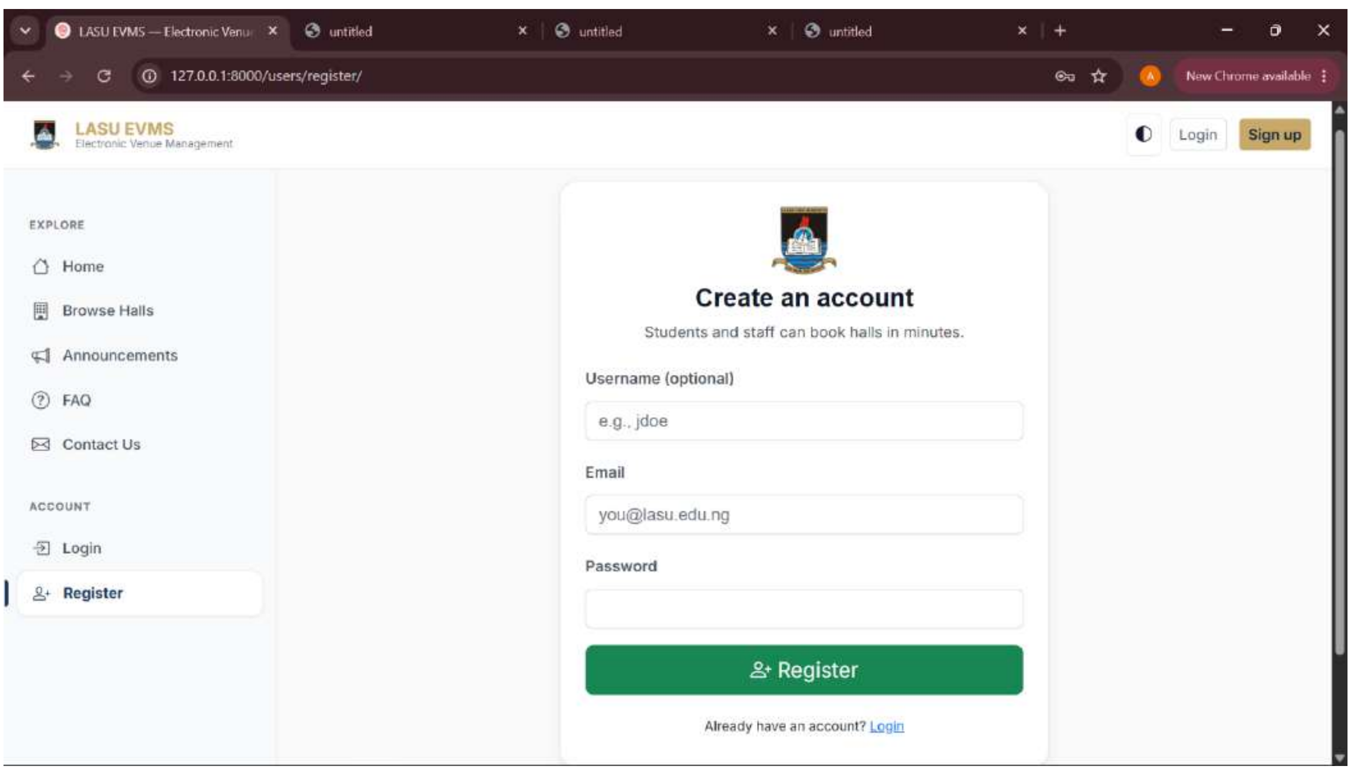
New Chrome available
LASU EVMS
Electronic Venue Management
Login
Sign up
EXPLORE
Home
Browse Halls
Announcements
FAQ
Contact Us
ACCOUNT
Login
Register
Contact Us
Have a question? Send us a message.
Name
Email
Subject
Message
Paragraph
Send message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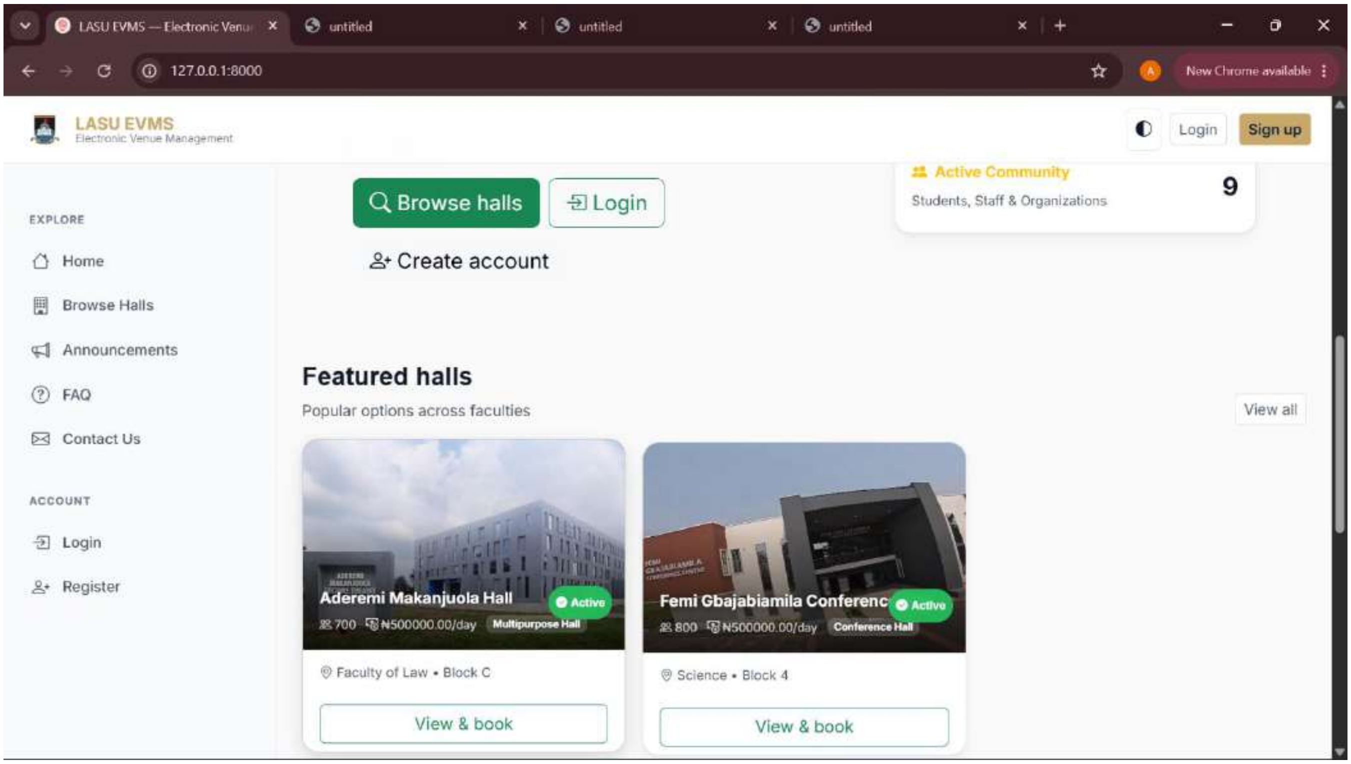
New Chrome available
LASU EVMS
Electronic Venue Management
Login
Sign up
EXPLORE
Home
Browse Halls
Announcements
FAQ
Contact Us
ACCOUNT
Login
Register
Create an account
Students and staff can book halls in minutes.
Username (optional)
e.g., jdoe
Email
you@lasu.edu.ng
Password
Register
Already have an account? Login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Electronic Venue Management
Login
Sign up
EXPLORE
Home
Browse Halls
Announcements
FAQ
Contact Us
ACCOUNT
Login
Register
Browse halls
Login
Create account
Active Community
Students, Staff & Organizations
9
Featured halls
Popular options across faculties
View all
Aderemi Makanjuola Hall
Active
700
N500000.00/day
Multipurpose Hall
Faculty of Law • Block C
View & book
Femi Gbajabiamila Conferenc
Active
800
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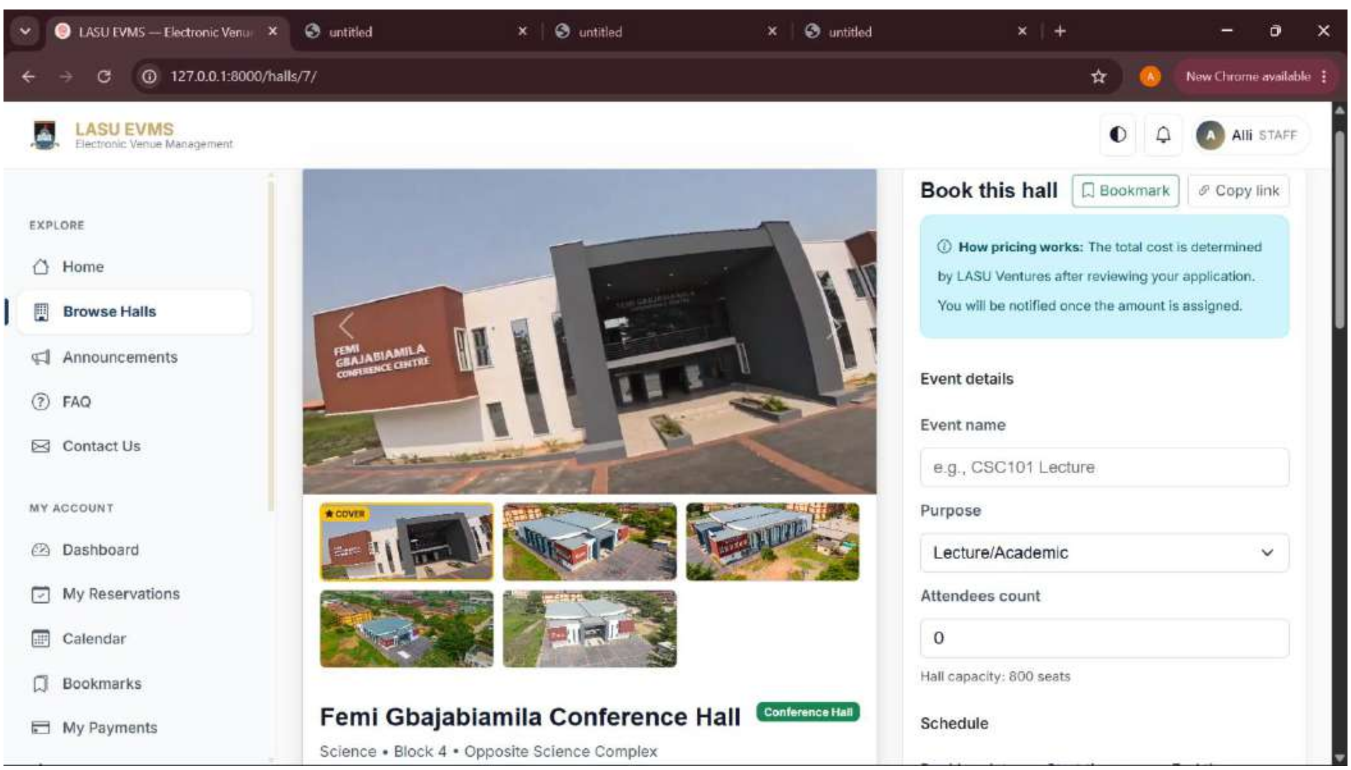
Conference Hall
Science • Block 4
View & book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New Chrome available
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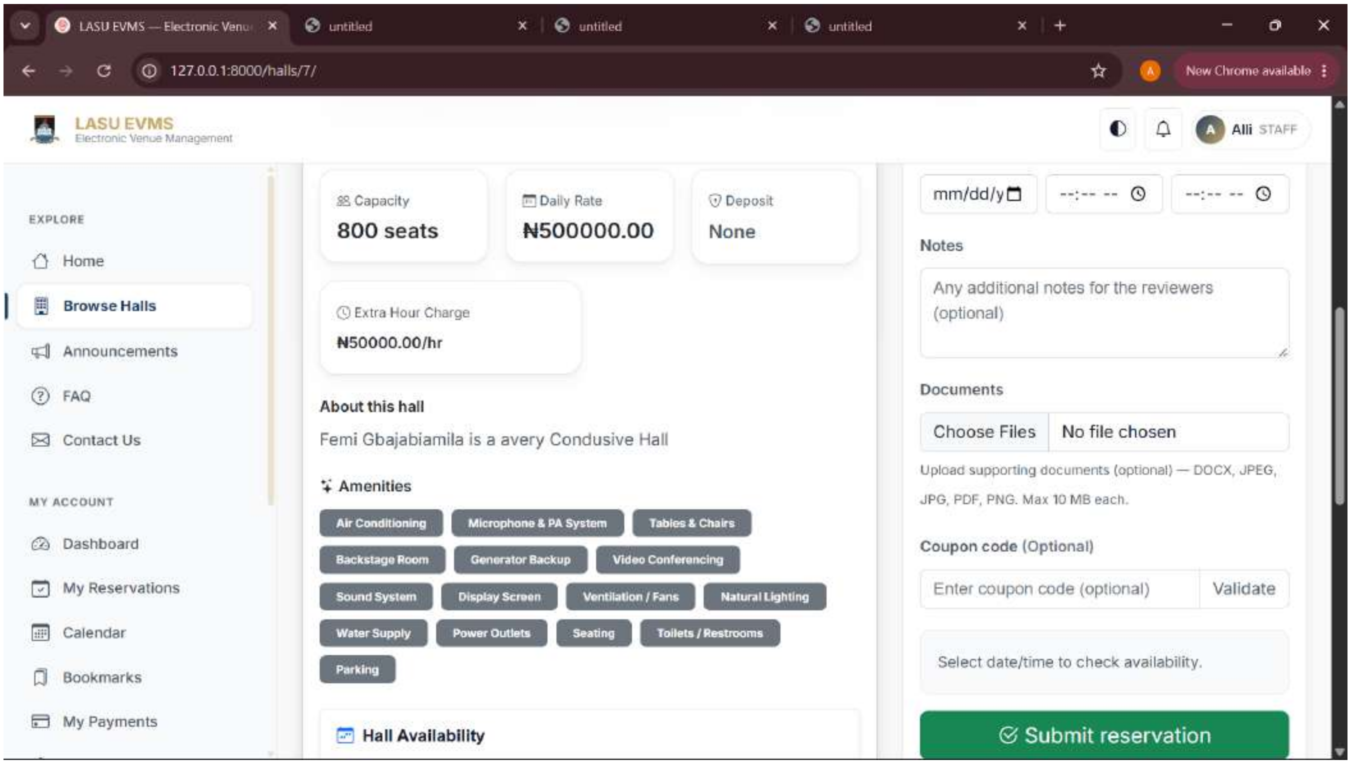
Electronic Venue Management
Alli STAFF
EXPLORE
Home
Browse Halls
Announcements
FAQ
Contact Us
MY ACCOUNT
Dashboard
My Reservations
Calendar
Bookmarks
My Payments
COVER
Femi Gbajabiamila Conference Hall
Conference Hall
Science • Block 4 • Opposite Science Complex
Book this hall
Bookmark
Copy link
How pricing works: The total cost is determined by LASU Ventures after reviewing your application. You will be notified once the amount is assigned.
Event details
Event name
e.g., CSC101 Lecture
Purpose
Lecture/Academic
Attendees count
0
Hall capacity: 800 seats
Schedule



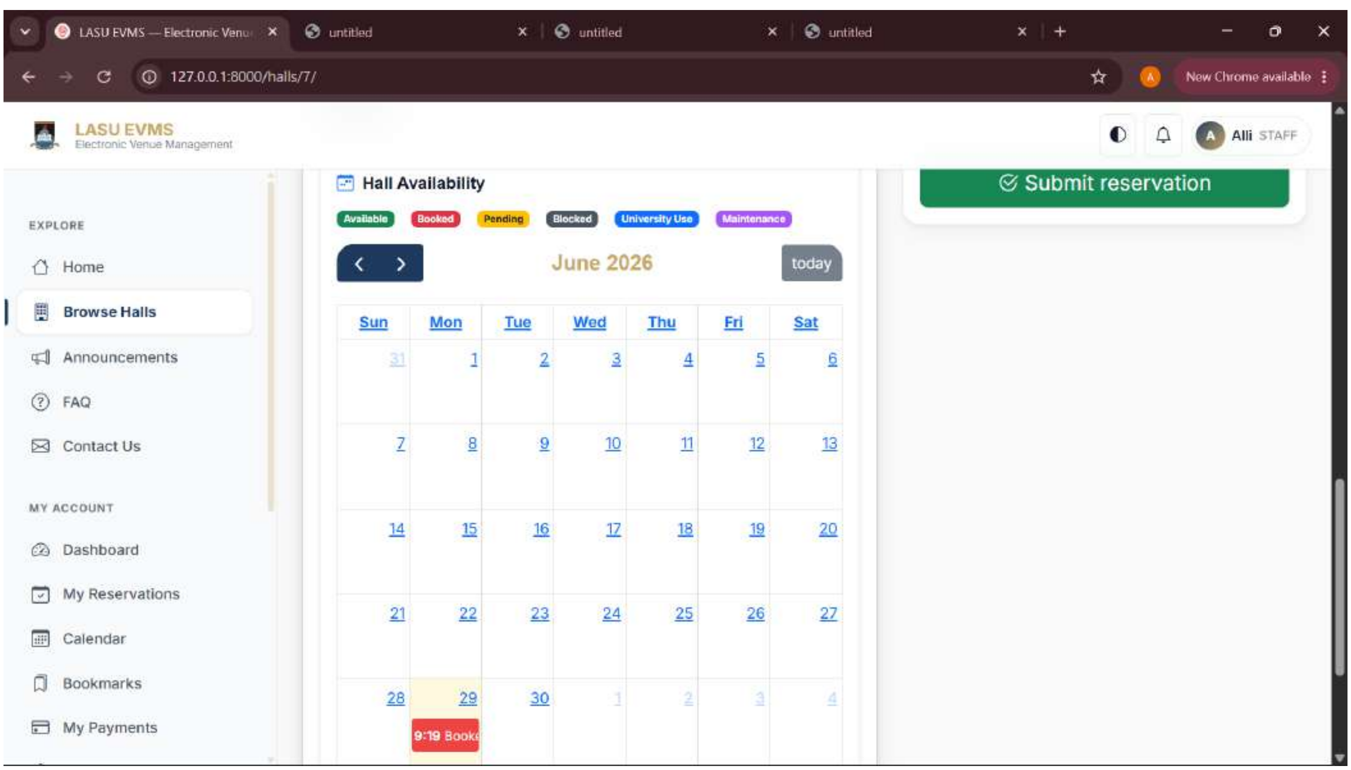


Figure 4.3: The public-facing facility catalog, Anouncements, contact us, login page, booking page and the real-time availability calendar.

CHAPTER FIVE

SUMMARY, CONCLUSION, AND RECOMMENDATIONS

5.1 Summary of the Project

The manual administration of event halls at Lagos State University historically generated profound logistical challenges, including scheduling conflicts, opaque approval timelines, and weak financial tracking. This project was undertaken to replace these fragmented processes with a cohesive, digital solution.

The resulting Event Hall Management System (EVMS) is a robust, web based application built upon the Django framework. Through careful requirement analysis and iterative development, the software successfully digitized the complex approval hierarchy of the university. It implemented a strict, state driven workflow that seamlessly coordinates the distinct responsibilities of the Ventures Unit, Facility Management, and the Bursary Unit. By incorporating algorithmic conflict detection, Role Based Access Control, automated financial verification queues, and post event damage accountability, the system provides an end to end management tool tailored precisely to the operational realities of the institution.

5.2 Conclusion

The deployment of the EVMS demonstrates that generic scheduling software is insufficient for large academic institutions. To achieve true operational efficiency, software must be structurally aligned with the internal bureaucracy of the institution.

By modeling the system around a tripartite authorization framework (commercial approval by Ventures, physical clearance by Facility Management, and financial verification by Bursary) the application eliminates the communication silos that previously plagued the reservation process. The mathematical enforcement of time slots ensures that double booking is eradicated. Furthermore, the integration of immutable audit logs and integrated penalty management transforms the booking process from a simple calendar exercise into a fully accountable, enterprise grade resource management operation.

In conclusion, this project provides Lagos State University with a powerful technological foundation capable of bringing transparency, speed, and reliability to its facility management operations.

5.3 Recommendations

To maximize the impact and longevity of the EVMS, the following strategic and technical recommendations are proposed:

1. **Production Database Migration:** The underlying SQLite database used during development must be migrated to a robust, highly concurrent database engine such as PostgreSQL prior to campus wide deployment. This is essential to maintain performance integrity when hundreds of students attempt to access the platform simultaneously.
2. **Implementation of Asynchronous Task Queues:** Currently, email notifications are processed synchronously. To improve the user experience and reduce page load times, the system should integrate an asynchronous task queue like Celery to handle email dispatching in the background.
3. **Physical Infrastructure Integration:** The logical security of the software can be extended into physical security by integrating the application with biometric or RFID door scanners. This would allow the system to automatically lock and unlock specific halls strictly according to the approved reservation schedule.
4. **Mandatory Institutional Training:** The success of the software depends on user adoption. The university administration should conduct mandatory training workshops for all Ventures, Facility, and Bursary personnel to ensure they are proficient in utilizing their respective digital dashboards, thereby accelerating the transition away from paper based routines.

5.4 Contribution to Knowledge

This research contributes a highly specific, context aware architectural model to the field of educational facility management in developing regions. While existing literature extensively covers automated scheduling, it rarely addresses the programmatic coordination of complex, multi departmental financial and physical approval hierarchies.

By detailing the exact database structures, state machine transitions, and software design patterns required to digitally synchronize a Ventures Unit, a Facility Management department, and a Bursary Unit, this project provides a reproducible software blueprint. It serves as a practical reference for other institutions seeking to build bespoke administrative software that respects, rather than ignores, their established internal governance structures.

5.5 Final Remarks

The Event Hall Management System represents a critical step forward in the modernization of internal operations at Lagos State University. It proves that by applying rigorous software engineering principles to real world administrative bottlenecks, institutions can vastly improve their service delivery, financial accountability, and operational transparency.

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